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Air transport efficiency and demand factors

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*To my family,
for their unconditional love, sacrifices, and unwavering support
throughout every step of my academic journey.*

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Abstract

In the modern economy, airports serve as hubs for trade and sustainability, as well as for connectivity between countries. This master thesis examines first the dynamic relationship between the determinants of passenger demand and the operational efficiency of 51 airports in Europe during the period 2014–2024.

The study refers to two subjects, first of all the study raises to the point of the efficiency of the airports and the factors which has an effect on the efficiency which is a very important topic for today's aviation industry which can easily help an airport's administration get an airport in a high business level. Also the fact that the efficiency of an airport can easily be effected by other macroeconomic, demographic and geographic variables can be considered as an index for their next business moves, which also is mentioned in this master thesis. The next subject which is presented in this master thesis is the passenger demand in aviation industry, this question is also an important topic for the industry because the wave of passengers is an important matter about how the infrastructures of the airport can handle fluctuating passenger demand.

The results of the three methods yielded significant conclusions regarding the topics under analysis. More specifically, the results of the first stage—that is, the demand model—showed that the main drivers of passenger demand are none other than the population and tourist arrivals, confirming that the European market is tourism-driven.

In the second stage of the analysis that is, the DEA efficiency analysis a sharp decline in efficiency was revealed during the pandemic period. What is interesting is that after the end of the pandemic specifically during the recovery period in 2024 it was observed that major hubs such as Eleftherios Venizelos Airport in Athens, Madrid, and Barcelona returned to the efficiency frontier as “shining stars.”

In short, the study fills an important research gap by providing up to date data on the post pandemic era and offering a comprehensive analysis that links demand to operational excellence at the pan-European level.

Key words: European Airports, Passenger Demand, Airport Efficiency, DEA Model, Tobit Model, Covid-19, Tourism

Abstract

Στην σύγχρονη οικονομία, τα αεροδρόμια λειτουργούν ως κόμβοι εμπορίου και βιωσιμότητας, καθώς και διασυνδεσιμότητας μεταξύ των χωρών. Η παρούσα διπλωματική εργασία εξετάζει αρχικά τη δυναμική σχέση μεταξύ των προσδιοριστικών παραγόντων της επιβατικής ζήτησης και της λειτουργικής αποτελεσματικότητας 51 αεροδρομίων στην Ευρώπη κατά την περίοδο 2014–2024.

Η μελέτη αναφέρεται σε δύο θέματα. Καταρχάς, αναδεικνύει το ζήτημα της αποτελεσματικότητας των αεροδρομίων και των παραγόντων που την επηρεάζουν, ένα ιδιαίτερα σημαντικό θέμα για τη σημερινή αεροπορική βιομηχανία, το οποίο μπορεί να βοηθήσει τη διοίκηση ενός αεροδρομίου να το οδηγήσει σε υψηλό επιχειρηματικό επίπεδο. Επίσης, το γεγονός ότι η αποτελεσματικότητα ενός αεροδρομίου μπορεί εύκολα να επηρεαστεί από άλλες μακροοικονομικές, δημογραφικές και γεωγραφικές μεταβλητές μπορεί να θεωρηθεί ως δείκτης για τις επόμενες επιχειρηματικές τους κινήσεις, κάτι που επίσης αναφέρεται σε αυτή τη διπλωματική εργασία. Το δεύτερο θέμα που παρουσιάζεται είναι η επιβατική ζήτηση στην αεροπορική βιομηχανία. Το ζήτημα αυτό είναι εξίσου σημαντικό, καθώς η επιβατική ροή αποτελεί κρίσιμο παράγοντα για το πώς οι υποδομές του αεροδρομίου μπορούν να διαχειριστούν τις διακυμάνσεις της ζήτησης.

Τα αποτελέσματα των τριών μεθόδων οδήγησαν σε σημαντικά συμπεράσματα σχετικά με τα υπό εξέταση θέματα. Ειδικότερα, τα αποτελέσματα του πρώτου σταδίου —δηλαδή του υποδείγματος ζήτησης— έδειξαν ότι οι κύριοι κινητήριοι μοχλοί της επιβατικής ζήτησης δεν είναι άλλοι από τον πληθυσμό και τις τουριστικές αφίξεις, επιβεβαιώνοντας ότι η ευρωπαϊκή αγορά είναι τουριστικά κατευθυνόμενη.

Στο δεύτερο στάδιο της ανάλυσης, δηλαδή στην ανάλυση αποτελεσματικότητας DEA, αποκαλύφθηκε μια απότομη πτώση της αποτελεσματικότητας κατά την περίοδο της πανδημίας. Το ενδιαφέρον είναι ότι μετά το τέλος της πανδημίας, και συγκεκριμένα κατά την περίοδο ανάκαμψης το 2024, παρατηρήθηκε ότι μεγάλοι κόμβοι, όπως το Αεροδρόμιο «Ελευθέριος Βενιζέλος» της Αθήνας, της Μαδρίτης και της Βαρκελώνης, επανήλθαν στο μέτωπο αποτελεσματικότητας ως «λαμπρά αστέρια» ("shining stars").

Συνοπτικά, η μελέτη καλύπτει ένα σημαντικό ερευνητικό κενό παρέχοντας επικαιροποιημένα δεδομένα για τη μεταπανδημική εποχή και προσφέροντας μια ολοκληρωμένη ανάλυση που συνδέει τη ζήτηση με τη λειτουργική αριστεία σε πανευρωπαϊκό επίπεδο.

Λέξεις-κλειδιά: Ευρωπαϊκά Αεροδρόμια, Επιβατική Ζήτηση, Αποτελεσματικότητα Αεροδρομίων, Μοντέλο DEA, Μοντέλο Tobit, Covid-19, Τουρισμός

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1 Introduction

Aviation and airports are an impressive sector that is considered to contribute to development both at the national level and to the global economy(Oum et al. (2005)). Literature sees airports as engines of growth because of the connectivity and tourist development, experiencing ups and downs due to the complexity of the variables that affect it (Oum et al. (2005),Adrián Nerja (2021),Kashiramka et al. (2016)). More specifically, as mentioned, they are considered critical catalysts for regional development, international trade, social cohesion, and tourism. This is due to the fact that, thanks to the continuous exchange of passengers and goods, businesses from different countries are able to expand their trade in other countries, potentially creating cooperation between countries, resulting in real growth and an increase in global GDP. At the same time, aviation results in the employment of thousands of workers, which in turn contributes to improving the living standards of the country(Senguttuvan (2006),Fragoudaki & Giokas (2016)).More specifically, due to the complex process that takes place from the moment the plane arrives until the moment it leaves again, there are positions like baggage loading and unloading, in-flight meals, fueling, and many more, It requires the staff to work not only on the same flight but also on many others and with different types of aircraft, which means thousands of employees are needed for the airport to operate efficiently(Fragoudaki & Giokas (2016)).

The main question here is how we can know when an airport is efficient, as well as the 'distance' it needs to reach a point in order to operate at a top level. In the literature, it is mentioned that airport efficiency can be measured using parametric and non-parametric methods (Yang (2010),Pels et al. (2003),Karanki & Bilotkach (2023), etc.), which take as variables the inputs and outputs of airports, such as the number of runways and terminals, as well as the number of arrivals and cargo(Gillen & Lall (1997), Pels et al. (2003), etc.). Other studies in the literature have shown that aside from these factors, which are thought to help an airport operate effectively with some potential development, there are also different factors that can indicate an increase in airport efficiency. More specifically, we're talking about macroeconomic factors that show the country's standard of living, as well as geographical and social factors Just like the passenger demand that exists from people(Tsekeris (2011) & Fernandes & Pacheco (2002),Kaya et al. (2022), etc.).

So, it can be asked which factors can increase the efficiency of airports, and importantly, not be factors related to the infrastructure and financial aspects of the airport as a business(Kaya et al. (2022),Yilmaz et al. (2022), etc.). It is impressive, therefore, that an airport can be considered not just a means of transportation, but a pillar of living standards, economic, and commercial development, and moreover, its efficiency as a business can be measured, which is influenced by various factors outside the business itself(Kashiramka et al. (2016) etc.). In this research, these questions will be answered

and some impressive findings will be presented.

As we mentioned before, the efficiency of airports is influenced by internal operational factors as well as external macroeconomic and other factors. An important factor that significantly affects airport efficiency is the passenger demand from the public. Specifically, demand is considered the reason why an airport is seen as a pillar of tourism development and possibly of the country and its GDP (Valdes (2015), Kluge, Paul, Cook & Cristóbal (2017)). Therefore, it is important for the management of such a pillar to know what really causes fluctuations in such an important factor. According to the literature, passenger demand is influenced by different macroeconomic, socio-demographic, and various other factors, making it a complex variable with many ups and downs due to the complexity of the variables that affect it. In this research, all these factors are combined and all the questions that were raised are answered, so that the way the industry operates is understandable and it can also be considered a useful guide for an airport manager to take the right steps with the sole aim of growth (Kaya, Aydın, Karadayı, Ülengin & Ülengin (2022)).

It is important to note that the research focuses on data from the European Union since it is considered a continent with many legislative changes in the sector as well as the creation of different business models. More specifically, Europe has historically faced issues of state interests being exploited through monopolies and blocking free competition (Sismanidou & Tarradellas (2017)). Thus, in the 1990s, the European Union eliminated this behavior through the liberalization process by breaking up monopolies and freeing competition through three legislative packages (Dobruszkes, 2006). As a result, legislation had to ensure free access to routes and time slots, as well as the right for cabotage, thus improving the productivity level of the sector (Suau-Sanchez et al., 2020) and the development of European countries.

In the continuation of the industry's story, it's important to note the creation of different business models. More specifically, we're talking about the Low-Cost model of business, and classic examples are the Ryanair group and easyJet (Dieu Huynh et al. (2025)). The presence of such business models caused problems for most airlines due to the increasing demand for these types of flights. So, the literature mentions that the demand for low-cost flights affected overall air travel demand, leading to a reshaping of the business strategies of other airlines and the development of a high level of competition (Fernández, Coto-Millán & Díaz-Medina (2018)).

The history of European aviation, apart from the things mentioned earlier, also includes the major trend of airport privatizations, as they turned from simple state public services into independent commercial enterprises operating based on business growth, profitability, and efficiency criteria (Oum, Yan & Yu (2008)). This trend progressed to concessions and privatizations of both large and small infrastructures, such as the UK's BAA (Parker (1999)) and Spain's AENA (Martín & Román (2001)). It's also important to

note that in our country, Greece, 14 regional Greek airports were handed over to Fraport Greece(Fragoudaki & Giokas (2020)).We can therefore conclude that the study of European aviation was not random but was considered interesting because of all the factors mentioned earlier, as it is regarded as a continent that has gone through major crises and changes in the industry(Scotti, Malighetti, Martini & Volta (2012), Pels, Nijkamp & Rietveld (2003)).

Aviation in general, aside from Europe and all the changes we mentioned earlier, has gone through major economic, geopolitical, and health crises that have caused significant difficulties both for the industry as a whole and for the companies within it(Gelhausen et al. (2013)). More specifically, the energy instability in the energy sector and the price of oil has been and still is a major expense for airlines(Felix & Saucedo (2025)). Geopolitical crises, like those in the Strait of Hormuz(Morgan (2026)), have led to sharp fluctuations in oil prices, which significantly affect fare prices, ultimately influencing passenger demand.

Another significant crisis that clearly leaves a mark on international aviation is none other than environmental restrictions and sustainability. More specifically, emissions, which are estimated at 2-3% of global carbon dioxide (CO₂)(Čubić & Török (2025)), enforce strict rules like the 'fit for 55' initiative, which is a comprehensive package of legislative proposals and policies by the European Union aimed at reducing net greenhouse gas emissions by at least 55% by 2030 compared to 1990 levels(Künnen et al. (2023)), making efficiency optimization and fuel waste reduction an urgent necessity.

The efficiency of airports is also noted to be significantly tested within the general framework of aviation in terms of infrastructure organization and business management(Senguttuvan, 2006; Jacquillat and Odoni, 2018). More specifically, the rapid increase in passenger demand encountered obstacles due to the insufficiency of infrastructure such as runways and terminal stations, resulting in a huge volume of flight delays, which leads to significant economic costs and compensation for passengers and airlines(Gillen et al., 2016; Karanki and Bilotkach, 2023). This issue is an important research outcome, as the results that could be found as a way to improve airport management could help avoid the downgrading of the industry due to the major impacts it can have on international development(Ahn and Min, 2014; Keček et al., 202x).

A modern problem faced both by international aviation and the global population is the years of the 2020-2022 crisis, and specifically the Covid-19 crisis(Gao, 2022; Dieu Huynh et al., 2025). More specifically, Covid-19 led to an unprecedented collapse in passenger traffic, which the literature refers to as a state of complete stagnation, meaning below 5% of 2019 levels during the lockdown(Gao, 2022). Their efficiency was also found to be in complete collapse, as airports suffered a dramatic drop in efficiency scores due to the inability to reduce their fixed operational and construction costs(Pels et al., 2003; Ripoll-Zarraga and Mar-Molinero, 2020; Keček et al., 202x).

The growth period came in 2023-2024, which marks the full return to normalcy,

with traffic and operating expenses historically exceeding pre-pandemic levels, testing the resilience of their infrastructure(Karanki and Bilotkach, 2023; Keček et al., 202x). This study also raises the question of how the pandemic affected both the performance of the sample airports and the demand for air travel, since it contains data from 2014 to 2024(Gao, 2022; Dieu Huynh et al., 2025).

So, to sum up from the above, the aviation sector needs research regarding the improvement of productivity and efficiency in how it handles the continuously increasing passenger traffic that is created in the global industry(SEO Amsterdam Economics, 2024; Kashiramka et al., 2016). In short, the main questions that are useful to develop and answer, which can reliably be used as a guide for the management of modern airports, are how modern crises affect their performance, that is, finding the efficiency of airports before, during, and after, drawing useful conclusions(Kaya et al., 2022; Keček et al., 202x). Once we have found efficiency, it is also useful to understand which other factors influence it, and more specifically, since the pandemic crisis certainly affected efficiency, but according to the literature, there are other factors that influence it, which makes it even more impressive that efficiency can both increase and decrease during periods of crisis, we can specifically understand which social-demographic, macroeconomic, geographical, and many other factors can help prevent the sector from being destroyed during such times(Tsekeris, 2011; Fragoudaki and Giokas, 2016; Yilmaz et al., 2022).

A common variable mentioned in the literature as a key factor for airport efficiency is people's demand for air travel(Pels et al., 2003; Ngo and Tsui, 2020). As expected, periods of crisis directly affect this variable for many different reasons, so it is considered important to find the factors that directly influence travel demand in air transport, as well as how this variable moves during periods of crisis, during, and after the crisis, that is, in recovery periods(Gao, 2022; Dieu Huynh et al., 2025; Avcı and Avcı, 2025).

Modern literature covers most questions, but there are considered to be significant gaps regarding the data used and the time span during which the studies take place. More specifically, most researchers choose to examine in fragments either the factors that affect flight demand or the way in which airport operational efficiency is determined (Pels et al., 2003; Fragoudaki and Giokas, 2016). So far, there have been very few attempts in the literature to connect these two fields under a single econometric framework that uses macroeconomic data to explain both demand and infrastructure inefficiency at the same time.

Also, a large portion of the literature takes place in periods before 2019, so there is a serious lack of literature regarding the Covid-19 pandemic period(Gao, 2022). As a result, there was a significant research gap in evaluating the resilience of airports during the pandemic crisis, but especially during the recovery period after 2022(Karanki and Bilotkach, 2023; Keček et al., 202x).

Another gap that exists in the international literature is the geopolitical scope of

the research. Specifically, the literature is developed in just one country(Tsekeris, 2011; Fragoudaki and Giokas, 2016; Yilmaz et al., 2022; Keček et al., 202x), so using data from multiple countries covers a wide range, from developed to less developed countries, and this way the research can include everything from major airports to small regional ones.

In this research, these gaps are covered since the data used spans a wide range of countries in the European Union and, as it continues, data from major passenger hubs like Munich Airport are used, as well as smaller regional airports like Mykonos Airport(Pels et al., 2003; Fragoudaki and Giokas, 2020). Also, because of the years covered by the research, namely from 2014 to 2024, the issue of the COVID-19 pandemic is addressed—that is, years before, during, and the recovery period(Gao, 2022; Keček et al., 202x).

2 Literature Review

In this chapter, we will provide the corresponding literature review. This master thesis, as it was said before, is about two main questions which concern aviation industry and mainly the airports businesses. More specifically, the literature review will provide conclusion from articles with similar topics, which are the airport efficiency and the factors which has effect on the efficiency and the factors of passenger demand.

Many articles, which are mentioned for these topics came across interesting conclusions even though it took place across different samples and more specifically in different countries and continents. In this research, we take we take 42 different articles into consideration which are taking place in different sample. That mean we get a global overview of the topic.

2.1 Airport Operational Efficiency and Exogenous Determinants

Airport efficiency and the factors which which one can consider as guides for it are a very popular topic in research for aviation industry(Kaya et al., 2022; Iyer & Jain, 2019; Yang, 2010). More specifically, efficiency is the main ability of an airport to maximize outputs given that it has some available incoming resources(Ahn & Min, 2014; Ripoll-Zarraga & Mar-Molinero, 2020) or differently or, alternatively, to minimize resource usage for achieving a specific level of service(Pels et al., 2003; Ahn & Min, 2014; Keček et al., 2026).

As it clearly understood, airport efficiency is a highly important index for the managementYang, 2010; Kaya et al., 2022; Keček et al., 2026) because knowing their level of the efficiency can rise the level of business operations to higher levels (Yang, 2010; Ahn & Min, 2014; Keček et al., 2026).

The literature demonstrates that that airport efficiency is not stable but is instead affected by many factors across different thematic areas, specifically within the economic (Carlucci et al., 2018; Kaya et al., 2022), geographic (Fragoudaki & Giokas, 2016; Pagoni & Koumoutsidi, 2022), and demographic fields (Kaya et al., 2022; Yilmaz et al., 2022).

Being more specific, numerous author have addressed one of the most prominent topics in the aviation industry which is the topic of privatization and ownership dynamics(Iyer & Jain, 2019; Kaya et al., 2022). More specific, in literature the impact of ownership remains highly controversial due to widely conflicting conclusions (Fernández et al., 2018; Gutierrez & Lozano, 2016; Iyer & Jain, 2019; Kaya et al., 2022; Scotti et al., 2012).

Table 1: Comprehensive Literature Synthesis: Airport Privatization, Ownership, and Institutional Performance

Author(s)	Year	Methodology	Key Findings & Conclusions
Bennett & Johnson	1980	Financial ratios & empirical cost analysis	Private firms systematically outperform SOEs in cost management and resource allocation (property rights theory).
Boyd	1986	Econometric cost-minimization analysis	Found no significant evidence that public enterprises are inherently less cost-efficient under similar market conditions.
Boardman & Vining	1989	Econometric performance analysis	Mixed enterprises perform worse than pure private or pure state-owned firms due to conflicting shareholder goals.
Ehrlich et al.	1994	Dynamic panel data estimation	Private ownership drives higher long-term productivity growth and technical progress compared to public/mixed firms.
Parker	1999	DEA on BAA (pre/post concession)	BAA privatization (1987) showed no immediate efficiency gains; ownership transfer requires market competition to perform.
Oum, Yan & Yu	2008	Bayesian Translog Frontier SFA Cost	Majority-private and independent public corporations achieve highest cost efficiency; direct government control causes inefficiencies.
Scotti et al.	2012	Multi-Output Translog (SFA) SDF	Privatization increased technical inefficiency (+1.0176) due to large capital investments without proportional short-term traffic growth.
Gutiérrez & Lozano	2016	Output DEA VRS & Fractional Regressions	Private ownership significantly improves operational efficiency (+0.3260 Tobit, +2.4826 Logit) in small/medium European airports.
Fragoudaki & Giokas	2020	DEA (CCR/VRS), MPI & Mann-Whitney U	Selection bias in Greece: Concessional regional airports (Fraport) had higher efficiency pre-privatization due to cherry-picked state assets.

Table 1: Comprehensive Literature Synthesis: Airport Privatization, Ownership, and Institutional Performance

Author(s)	Year	Methodology	Key Findings & Conclusions
Kaya et al.	2022	Super-SBM DEA & Simar-Wilson Reg.	Private concessions had a negative efficiency effect (-1.81), driven by excess terminal capacity and infrastructure overinvestment.

The table above summarizes the literature addressing airport ownership and its impact on performance. As is evident the ownership of an airport is highly debated and far from undeniably positive. While some researchers find that private ownership or privatization enhances operational efficiency (Barros & Dieke, 2007; Oum et al., 2006; lo Storto, 2018) , others conclude that it has no noticeable impact (Parker, 1999; Lin & Hong, 2006), or even observe that publicly owned airports outperform their private counterparts (Gutierrez & Lozano, 2016; Scotti et al., 2012).

From a theoretical perspective, property rights theory and agency theory argue that the transfer of control to the private sector introduces superior managerial incentives, stricter fiscal discipline, and a clear orientation towards profit maximization.

Bennett and Johnson (1980) and Ehrlich et al. (1994) provided the earliest findings demonstrating that private businesses present better yields and faster productivity growth compared to state-owned enterprises. In a similar vein, Oum, Yan, and Yu (2008) applied this theory to the aviation industry; using a Stochastic Frontier Analysis (SFA) translog cost function on a sample of 108 airports, they concluded that most privately owned airports and airports under public corporate management succeed in achieving optimal cost efficiency. On the contrary, direct state control or joint management by multiple levels of local government leads to serious inefficiencies. However, this link between private initiative and operational improvement was not supported by Gutiérrez and Lozano (2016) regarding European small and medium-sized airports (SMAs), where public ownership contrary to their expectations consistently demonstrated a positive and statistically significant effect on operational efficiency.

Conversely, a substantial body of literature challenges the necessity of private capital for efficiency optimization, adopting a more skeptical or neutral stance. Boyd (1986) challenged the property rights consensus, finding no systematic or statistically significant empirical evidence that public enterprises are inherently less cost-efficient than private firms when operating under similar market conditions. This is empirically validated in airport operations by Parker (1999), who investigated the landmark privatization of the British Airport Authority (BAA) in 1987 using DEA. Parker concluded that the transfer to private ownership did not lead to immediate or notable technical efficiency

improvements, suggesting that ownership changes alone—without competitive pressure or regulatory reforms—are insufficient to drive efficiency gains.

Further challenging the privatization hypothesis, some researchers identify statistically significant negative impacts associated with private sector involvement and mixed public-private ownership. Boardman and Vining (1989) demonstrated that mixed enterprises (where the state and the private sector share ownership) often perform worse than both pure private and pure public firms, due to conflicting organizational objectives and managerial friction between public and private shareholders. In airport-specific studies, Scotti et al. (2012) examined 38 Italian airports using a multi-output stochastic distance function and found that privatization statistically increased technical inefficiency. They argued that privatized operators in Italy tended to make large, capital-intensive, and indivisible investments (such as terminal expansions) that did not immediately translate into proportional traffic growth in the short run, thereby lowering technical efficiency scores. This negative effect is corroborated by Kaya et al. (2022) for the Turkish airport system, whose second-stage bootstrapped regression revealed a statistically significant negative coefficient for private property terminal operations, which they attributed to excess capacity and asset underutilization.

Finally, Fragoudaki and Giokas (2020) introduced a critical methodological dimension to this debate by identifying the presence of selection bias in the concession process. Studying the Greek regional airport system, they demonstrated that the 14 regional airports concessioned to Fraport Greece already exhibited statistically significant higher technical efficiency and productivity (TFP) indicators prior to their privatization. This suggests that the observed superiority of privatized airports is not necessarily a post-concession outcome of private management, but rather a result of state selection bias, as governments systematically choose the most lucrative, tourism-dense, and structurally efficient assets to offer to private investors.

Moving on to the next topic addressed in the literature, which concerns the size of the airport and its relationship to economies of scale. More specifically, airport size, namely the capacity or volume of passenger and cargo traffic handled, and its relationship with airport scale efficiency, which is the indicator demonstrating the extent to which an airport operates at its optimal operational scale.

Table 2: Literature Synthesis: Airport Size, Scale Efficiencies, and Economies of Scale

Author(s)	Year	Methodology	Key Findings & Conclusions
Murillo-Melchor	1999	DEA & Malmquist Index (33 Spanish airports)	Clear positive size-efficiency link; larger Spanish airports with higher passenger volumes show systematically higher technical efficiency.

Table 2: Literature Synthesis: Airport Size, Scale Efficiencies, and Economies of Scale

Author(s)		Year	Methodology	Key Findings & Conclusions
Bazargan & Vasigh		2003	DEA (45 US commercial airports)	Negative correlation between airport size and technical efficiency; smaller regional airports can operate more efficiently.
Pels et al.		2003	Joint SFA/DEA Translog frontiers (33 EU airports)	CRS for air transport movements (ATM), IRS for passenger movements (APM). Strong negative correlation (-0.83) between scale elasticity and size.
Martín et al.		2009–11	SFA cost frontiers (Spanish airports)	Larger airports enjoy higher operating/economic efficiency, confirming significant scale economies and lower marginal costs.
Tsekeris		2011	Output CRS/VRS DEA (39 Greek regional airports)	Scale is a major efficiency driver (24 airports $> 80\%$ scale efficiency); 27 out of 39 airports operate under IRS, favoring long-run expansion.
Lozano & Gutiérrez		2011	SBM-DEA & Malmquist Index (39 Spanish airports)	Larger airports show higher technical and scale efficiency; low-cost carrier (LCC) traffic significantly boosts scale performance.
Fragoudaki & Giokas		2016	Output DEA VRS & Bootstrap (38 Greek airports)	VRS isolates pure technical efficiency; second-stage regression confirms airport size (WLU) as a significant positive efficiency driver.
Fragoudaki & Giokas		2020	Output DEA CCR & Malmquist (Greek airports)	Large airports ($> 1\text{M}$ passengers) achieved highest technical efficiency (93.2% in 2015) and highest productivity growth (78%).
Yilmaz et al.		2022	Spherical Fuzzy AHP, DEA-BCC & Tobit (46 Turkish airports)	90.8% of Turkish airports operate under IRS due to inefficiently small scale; larger regional demand significantly boosts efficiency.
Karanki & Bilotkach		2023	Johansen SBM-DEA & Probit (59 US airports)	Large hubs show 9% higher capacity utilization than medium hubs; high-density traffic reduces excess capacity risks.

Table 2: Literature Synthesis: Airport Size, Scale Efficiencies, and Economies of Scale

Author(s)	Year	Methodology		Key Findings & Conclusions
Keček et al.	2023	Dynamic Window	DEA BCC (7 Croatian airports)	Large hubs face operational recovery frictions, while small regional airports maintain high economic efficiency via lean cost control.

Another primary structural debate in the literature concerns whether airport size and scale of operations dictate technical efficiency. A large portion of early empirical work, including studies by Coto-Millán et al. (2014) and Martín et al. (2009), validated the hypothesis that larger airports (major hubs) perform better than smaller ones due to the exploitation of significant economies of scale and density.

In a comprehensive study of European airports, Pels, Nijkamp, and Rietveld (2003) confirmed that while the average European airport operates under constant returns to scale (CRS) when handling air transport movements (ATM), it operates under strong increasing returns to scale (IRS) when moving passengers (APM), with smaller regional airports exhibiting the highest potential for scale-driven efficiency gains. This is strongly supported by Tsekeris (2011) for the Greek regional airport system, where the vast majority of peripheral airports were found to operate under IRS, justifying public policy recommendations for scale expansion and infrastructure upgrading.

However, the "size-efficiency" consensus is far from universal. Bazargan and Vasigh (2003) evaluated major US airports and identified a negative correlation between airport size and technical efficiency, demonstrating that smaller regional airports can achieve superior resource utilization. This is conceptually aligned with the findings of Ripoll-Zárpóga and Mar-Molinero (2020), who employed multivariate visualization to show that while large Spanish hubs achieve efficiency through massive passenger volumes, small regional airports can also achieve 100% technical efficiency on the frontier by specializing in alternative niche activities such as cargo or high punctuality.

Additionally, Yilmaz et al. (2022) raised the issue of "overinvestment" in regional infrastructures; they found that over 90% of Turkish regional airports operate under IRS with very low efficiency scores, suggesting that massive capital investments in terminal buildings and runways are often not matched by sufficient local tourist demand, leading to long-term asset underutilization.

In general economic theory, competition is considered the main driver for improving productivity, reducing operating costs and eliminating managerial inefficiency. However, in the aviation industry, the impact of spatial and local competition on the technical efficiency of airports is one of the most controversial and paradoxical issues in the international literature.

On the one hand, the liberalization of the aviation market and the strengthening of competition between airports in large national markets (such as the USA and China) act as a strong incentive for managers, forcing them to optimize the use of their physical assets and adopt technological innovations.

On the other hand, when competition occurs between neighboring airports sharing the same geographical area of influence (overlapping catchment area), empirical findings reveal that local competition can statistically significantly increase technical inefficiency. Because the basic infrastructure of airports (runways, terminals) are indivisible and require huge capital, neighboring airports often engage in destructive non-price competition, such as over-scheduling and parallel infrastructure investments, leading to overcapacity and under-employment of fixed assets.

Table 3: Literature Synthesis: Spatial Competition, Catchment Areas, and Airport Inefficiency

Author(s) & Year	Region	Methodology	Key Findings & Impact of Competition
Sarkis (2000)	USA	DEA (Operational efficiency)	Competitive pressure acts as a strong managerial catalyst for asset optimization and operational efficiency.
Chi-Lok & Zhang (2009)	China	Two-stage DEA & panel regressions	Liberalization and market competition significantly enhanced airport TFP and technological progress.
Martín et al. (2011)	Spain	Translog Stochastic Cost Frontier (SFA)	Airports in overlapping catchment areas are systematically cost-inefficient due to demand splitting, unless severely congested.
Scotti et al. (2012)	Italy	Multi-Output Translog SDF (SFA)	Local competition within drive-time catchment areas significantly increases technical inefficiency (+3.4519) via redundant capacity.
Merkert & Mangia (2014)	Italy / Norway	Two-stage DEA & truncated regressions	Intermodal and inter-airport competition dictates cost efficiency, with remote airports facing structural constraints.
Ripoll-Zárraga & Mar-Molinero (2020)	Spain	Multidimensional DEA & Cluster Analysis	Centralized management prevents price competition, forcing nearby regional airports to split stagnant local demand.

Table 3: Literature Synthesis: Spatial Competition, Catchment Areas, and Airport Inefficiency

Author(s) & Year	Region	Methodology	Key Findings & Impact of Competition
Nerja (2021)	Spain	Multi-output SFA & DEA models	Spatial competition within a 150 km radius reduces technical efficiency under centralized network models by splitting existing traffic.

While economic theory generally associates competitive pressure with improved productive performance, market discipline, and cost minimization, the empirical literature on airport economics reveals highly counter-intuitive and divergent findings regarding the role of spatial and local competition. In aviation markets, the impact of competition depends fundamentally on whether it operates at a national macro-level or between neighboring regional facilities sharing overlapping geographic catchment areas.

In large, liberalized aviation sectors, early empirical work validated the classical pro-competitive hypothesis. Sarkis (2000) evaluated major US commercial airports using Data Envelopment Analysis (DEA), concluding that competitive pressure among US hubs acts as a powerful managerially-driven catalyst for asset optimization, operational efficiency, and service quality improvements⁶. Similarly, in the context of emerging market deregulation, Chi-Lok and Zhang (2009) documented that market liberalization, airport localization programs, and intensified airline and airport competition in China significantly enhanced Total Factor Productivity (TFP) growth and stimulated technological progress across the airport network.

In stark contrast, when competition manifests between airports located in close geographic proximity—sharing the same local catchment area—empirical evidence from European airport systems demonstrates that competitive pressure frequently acts as a driver of technical and cost inefficiency. Scotti et al. (2012) conducted a multi-output stochastic distance function (SFA) analysis on 38 Italian airports, incorporating a drive-time potential demand competition index (CI). They proved that intense local competition among nearby airports had a statistically significant negative impact on technical efficiency ($CI = +3.4519^{***}$, where a positive coefficient in the inefficiency term indicates greater inefficiency)³⁴. Scotti et al. (2012) explained that competing airports frequently engage in wasteful "over-scheduling" and redundant, capital-intensive infrastructure expansions (such as passenger terminals and aprons) to attract airlines (particularly Low-Cost Carriers), leading to severe structural overcapacity and underutilized physical inputs. This finding is aligned with Oum, Yan, and Yu (2008), who demonstrated that the presence of multiple airports in a metropolitan region significantly increases cost

inefficiency for government-owned airports.

This phenomenon of "spatial inefficiency" is particularly pronounced in centralized network management models, such as Spain's AENA network. As analyzed by Mart'in, Rom'an, and Voltes-Dorta (2011), Ripoll-Z'arraga and Mar-Molinero (2020), and Nerja (2021), Spain maintains a centralized system where airports do not possess individual pricing autonomy and cannot compete via user charges or commercial discounts. Consequently, when two or more regional airports share a catchment area (within a 130–150 km radius), they cannot engage in price-based competition to expand the total travel market⁵¹⁶. Instead, they are forced to simply "split" or "fragment" the existing, stagnant local passenger demand. This demand splitting prevents regional facilities from achieving the traffic threshold necessary to capture economies of scale, resulting in widespread capacity underutilization and lower technical efficiency scores—particularly for interior regional airports that lack the natural competitive advantage of coastal tourism.

In the literature, it is generally posited that competition within the aviation sector can be driven by airline business models. More specifically, the emergence of low-cost carriers (LCCs) has transformed the aviation industry in terms of market competition, as well as airport efficiency.

Table 4: Comprehensive Literature Synthesis: Low-Cost Carriers (LCCs) and Airline Market Structure in Airport Efficiency

Author(s) Year	& Region / Context	Methodology	Key Findings & Impact on Airport Efficiency
Ha et al. (2010, 2013)	Northeast Asia	Two-stage DEA & Panel Econometrics	Uncovered an inverse U-shaped relationship between airline market concentration and airport technical efficiency, highlighting the role of buyer bargaining power.
Pyrialakou et al. (2012)	Regional Airports	Data Envelopment Analysis (DEA)	Found that high LCC penetration significantly alters airport resource utilization and directly enhances operational performance.
Coto-Millán et al. (2014)	Spain (35 AENA Airports)	Output-Oriented DEA & Truncated Regression	Showed that LCC presence strongly boosts airport scale efficiency by sustaining high passenger traffic volumes.

The table above summarizes the literature regarding the impact of low-cost carriers (LCCs) on airport efficiency. According to the literature, Coto-Millán et al. (2014)

demonstrate that the presence of LCCs enhanced scale efficiency by offering stable and high passenger volumes. This aligns fully with the study by Pyrialakou et al. (2012), who examined regional airports and concluded that LCCs exhibit high penetration in the aviation sector.

Furthermore, it is worth noting that Ha et al. (2010/2013) generally align with the aforementioned studies, demonstrating that airline market concentration (Airline Market Structure) impacts both the bargaining power and the efficiency of airports.

In the context of crises and external shocks, the literature indicates that crisis periods are characterized by high volatility, as an unexpected crisis can disrupt both the operational efficiency and overall functionality of airports.

Table 5: Comprehensive Literature Synthesis: Exogenous Shocks, Macroeconomic Determinants, and Crisis Resilience in Aviation

Author(s) & Year	Region / Context	Methodology	Key Findings & Impact of External Shocks & Macro Factors
Fragoudaki & Giokas (2020)	Greece (38 Airports)	Output-Oriented DEA, Malmquist Productivity Index (MPI) & Mann-Whitney U test	Proved that despite the severe Greek sovereign debt crisis, airport technical efficiency improved (+38.4% TFP) due to explosive incoming tourism and wage deflation, showing that strong external tourist demand can offset domestic macroeconomic shocks.

Table 5: Comprehensive Literature Synthesis: Exogenous Shocks, Macroeconomic Determinants, and Crisis Resilience in Aviation

Author(s) & Year	Region / Context	Methodology	Key Findings & Impact of External Shocks & Macro Factors
Gao (2022) / Suau-Sanchez et al. (2020)	Global / USA	Econometric time-series & scenario-based shock modeling	Characterized the COVID-19 pandemic as an unprecedented state of “complete stagna- tion” (traffic collapsing below 5% of baseline), causing a severe plunge in technical efficiency due to fixed infrastruc- ture cost rigidity and asset underutilization.
Keček et al. (202x)	Croatia (7 Airports, 2019–2023)	Dynamic DEA Window Analysis (BCC) under VRS	Revealed that COVID- 19 caused severe structural cost rigidity and efficiency plunges in major interna- tional hubs, whereas smaller regional air- ports demonstrated higher operational flexibility and faster post-crisis recovery.

Table 5: Comprehensive Literature Synthesis: Exogenous Shocks, Macroeconomic Determinants, and Crisis Resilience in Aviation

Author(s) & Year	Region / Context	Methodology	Key Findings & Impact of External Shocks & Macro Factors
Dieu Huynh et al. (2025)	Global Panel (52 Countries, 2010–2022)	Two-way Fixed Effects, 2SLS & Two-step Sys- tem GMM	Confirmed that COVID-19 caused a massive, statistically significant collapse in international tourist arrivals (coefficients −0.50 to −2.32), while accommodation ca- pacity and tourism labor markets displayed structural resilience during post-crisis recovery.

The table above provides a summary of the literature regarding the impact of crises on airport efficiency. More specifically, Fragoudaki & Giokas (2020), examining a sample of 38 Greek airports during the Greek economic crisis (2011–2015), report that a significant surge in international tourism combined with payroll cost containment allowed Greek airports to increase their productivity by 38.4%, thereby offsetting the domestic recession.

In line with these findings, Gao (2022) and Dieu Huynh et al. (2025) note that the pandemic caused a complete standstill, leading to a collapse in DEA efficiency scores, as airports were unable to scale down their indivisible physical infrastructure, resulting in substantial excess capacity.

An important theme in the literature concerns environmental ethics and the environmental problems that an airport can generate. The table below summarizes the key studies and their respective conclusions.

Table 6: Comprehensive Literature Synthesis: Undesirable Outputs, Flight Delays, and Environmental Efficiency in Airport Benchmarking

Author(s) & Year	Region / Context	Methodology	Key Findings & Environmental / Delay Integration
Yu (2004)	Taiwan (14 domestic airports, 1994–2000)	Directional Output Distance Function DEA with weak disposability	Introduced aircraft noise as an undesirable output and local population as an environmental variable; demonstrated that ignoring noise and environmental factors artificially inflates inefficiency and severely distorts airport efficiency rankings.
Diana (2010)	USA (New York area: JFK, EWR, LGA)	Stochastic Frontier Model (SFM) & VAR Granger Causality	Incorporated operational delays (taxi-out, block, airborne, taxi-in, gate arrival) into an SFM framework, successfully isolating true technical inefficiency from random operational noise and weather shocks.
Fan, Wu & Zhou (2014)	China (20 major airports, 2006–2009)	Directional Output Distance Function DEA with undesirable outputs	Integrated flight delays as an undesirable output in a joint production framework, revealing that accounting for congestion prevents delay-heavy hub airports from being unfairly penalized in technical efficiency evaluations.

Table 6: Comprehensive Literature Synthesis: Undesirable Outputs, Flight Delays, and Environmental Efficiency in Airport Benchmarking

Author(s) & Year	Region / Context	Methodology	Key Findings & Environmental / Delay Integration
Krstić Simić & Babić (2015)	Serbia / Europe (Air Traffic Management)	Environmental Inefficiency Metrics (IM) & Dynamic Complexity	Modeled Air Traffic Management (ATM) efficiency by evaluating fuel consumption (fuel inefficiency) and gas emissions (CO_2 , NO_x , HC, SO_x , H_2O , benzene), establishing the direct link between airfield layout tactics and environmental performance.

An important theme in the literature concerns environmental ethics and the environmental problems that an airport can generate. The table below summarizes the key studies and their respective conclusions. The integration of environmental impacts into the study of airport efficiency is deemed essential, as omitting undesirable outputs distorts performance evaluation results. Yu (2004) introduced aircraft noise as an undesirable output and local population as an environmental variable within a Directional Distance Function model, proving that their omission inflates inefficiency and distorts true airport rankings. Similarly, Krstić Simić & Babić (2015) modeled the efficiency of Air Traffic Management (ATM), linking airport operational tactics directly to excess fuel consumption and pollutant emissions, such as CO_2 and NO_2 .

Furthermore, treating flight delays as negative outputs allows for a fairer and more accurate measurement of technical efficiency. Diana (2010) incorporated various types of delays (such as taxi-out, airborne, taxi-in, and gate arrival) into a Stochastic Frontier Model (SFM), successfully isolating true technical inefficiency from random operational noise. In the same vein, Fan et al. (2014) embedded flight delays as an undesirable output into a Directional Distance Function model, demonstrating that accounting for congestion prevents the unfair underestimation of efficiency scores for large hub airports.

2.2 Airtraffic Passenger Demand

In this subchapter, we will refer to the literature related to passenger demand since, as we understood from the previous chapter, the issue of demand is an important factor in airport efficiency, as it provides a kind of forecast for the business strategy the airport

needs to follow as a business to be able to cope with the expected traffic. The table below summarizes the literature on demand.

Table 7: Literature Synthesis: Passenger Demand Models and Determinants

Authors Year	&	Methodology	Variables	Main Empirical Findings
Valdes (2015) [13]		Fixed Effects & Dynamic GMM	Dep: $\ln(Pax)$ Indep: GDP_{pc} , FDI , CPI , Ex , $Fuel$, LCC , $SARS$	Fixed Effects preferred ($\chi^2 = 58.01$). High income elasticity (> 1). GDP per capita, foreign investment, and LCC presence positively drive demand, whereas CPI exerts a negative effect. Strong dynamic demand persistence ($\beta = 0.284^{**}$).
Avci & Avci (2025) [20]		ARDL Granger Causality	Dep: Dom. Pax Indep: CPI , Ex , GDP	Cointegration confirmed ($F = 9.350$). Long-run demand is positively driven by GDP and CPI, while short-run CPI increases significantly depress domestic passenger traffic.
Dieu Huynh et al. (2025) [?]		FE, 2SLS & System GMM	Dep: ITA , $ROOM$, EMP Indep: LCC , GDP , EX , GOV , $COVID$	LCC expansion significantly boosts in- ternational tourist arrivals and hotel room occupancy. COVID-19 caused a severe structural decline in overall de- mand.
Felix Saucedo (2025) [?]		VECM Granger Causality	Endog: PAX_{LC} , PAX_{FS} Indep: $IGAE$, $INPC$, $TIIE$, WTI	Economic activity, inflation, and oil prices Granger-cause low-cost carrier passenger traffic, whereas full-service carrier demand is primarily driven by inflation.
Kluge et al. (2017) [?]		Robust OLS	Dep: $airtrips_{cap}$ Indep: $\ln(GDP_{pc})$, geo , $educ$, $urban$	Real GDP per capita and higher ter- tiary education levels ($\beta = 0.0022^{**}$) significantly increase per-capita air travel propensity across regions.

According to the literature, passenger air travel demand is a complex and multifaceted dynamic determined by an interplay of macroeconomic, monetary, structural, and socio-demographic factors. Rather than evaluating individual studies in isolation, the empirical evidence can be synthesized around key functional determinants of demand.

Economic expansion and rising national income serve as the primary long-run engine of air travel demand. Applying dynamic panel GMM and fixed-effects models across middle-income countries, Valdes (2015) demonstrated that real GDP per capita is the single most dominant long-term driver of demand, accounting for 75% of historical traffic growth. This long-run equilibrium relationship between economic growth and air passenger demand is empirically corroborated by Avcı & Avcı (2025) using ARDL bounds testing ($F = 9.350$) in Turkey, where a unidirectional Granger-causal flow running from GDP to passenger demand was established. Adding a socio-demographic dimension to income effects, Kluge et al. (2017) confirmed via robust OLS regression that per-capita GDP in PPP terms, alongside higher tertiary education levels ($\beta = 0.0022^{**}$), significantly enhances overall travel propensity.

While long-run traffic expansion is income-driven, short-run demand fluctuations are heavily conditioned by monetary shocks, price levels, and cost-side indicators. Examining domestic traffic in Mexico via Vector Error Correction Models (VECM) and Granger causality tests, Felix & Saucedo (2025) proved that Low-Cost Carriers (LCCs) are uniquely sensitive to monetary shocks, interest rates (TIE), fuel prices (WTI), and inflation (INPC), whereas Full-Service Carriers (FSCs) demonstrate higher demand resilience due to their corporate customer base. Short-run cost volatility is also addressed by Valdes (2015) and Avcı & Avcı (2025), who noted that while Consumer Price Index (CPI) inflation spikes temporarily reduce passenger purchasing power, long-run economic growth restores traffic equilibrium.

Structural innovations in airline business models and market deregulation policies act as major catalysts for travel demand and regional tourism development. Valdes (2015) found that Open Skies agreements with the European Union (OSEU) and Foreign Direct Investment (FDI) exert a statistically significant positive effect on passenger volumes. Expanding this framework to a global panel of 52 countries using IV/2SLS and System GMM estimators, Dieu Huynh et al. (2025) demonstrated that LCC seat capacity acts as a vital catalyst for international tourist arrivals, which subsequently stimulates accommodation supply (hotel rooms) and tourism employment. However, this growth trajectory remains vulnerable to severe black-swan events; both Valdes (2015) (for the 2003 SARS outbreak) and Dieu Huynh et al. (2025) (for COVID-19, reporting negative coefficients ranging from -0.500 to -0.927) documented acute traffic contractions during global health emergencies.

Finally, air travel demand is structurally anchored by spatial and demographic characteristics that define an airport’s catchment area. As shown by Kluge et al. (2017),

factors such as total population, urbanization rates, and insularity play a decisive role in shaping passenger traffic. Insularity, in particular, creates a structural reliance on air transport due to the absence of surface transport substitutes, significantly inflating per-capita air travel propensity in island regions.

3 Data and variables

3.1 Data Collection and Primary Sources

The sample used in this Master’s thesis was derived from a panel dataset compiled from multiple data sources, covering both the operational efficiency of airports and passenger demand dynamics for the period 2014–2024 for efficiency and 2018–2024 for passenger demand. The empirical framework incorporates operational metrics, macroeconomic determinants, regional demographic indicators, and institutional characteristics.

Eurostat served as a key reliable data source for several observations. More specifically, for the output variables of the DEA efficiency measurement model—namely Air Transport Movements (ATM) and passenger volume—data were retrieved from Eurostat’s aviation transport database (*avia_pa*). The Work Load Units (WLU) variable—a composite metric combining passenger volume and cargo tonnage handled by each airport—was also derived from Eurostat. This variable was utilized to address the strong correlation between these two components.

Furthermore, socio-demographic and regional variables were sourced from Eurostat, specifically total population, regional accommodation capacity, and educational attainment levels. Economic variables including the Consumer Price Index (CPI), Purchasing Power Parities (PPP), and energy-related metrics (fuel consumption) were likewise extracted from the same organization.

The World Bank also played a significant role as a highly credible source. Specifically, Real GDP per capita and foreign exchange rates were collected to capture national purchasing power and macro-level income effects. Additionally, variables such as international tourist arrivals and urban population were extracted to capture international demand flows and urbanization density.

In addition, because several input variables required for the DEA model were not available in public institutional databases, they were thoroughly investigated and extracted directly from individual airport annual reports spanning 2014 to 2024. Specifically, infrastructure variables (number of runways and passenger terminals) as well as financial metrics (operating revenues and staff costs) were hand-collected and standardized from audited financial statements and official annual reports of the respective airport operating authorities.

Finally, geographic and contextual controls were included as hand-constructed indi-

cators. A dummy variable assigning a value of 1 for island/isolated regional airports and 0 otherwise was constructed using GIS geographic coordinates and spatial mapping. Lastly, a COVID-19 pandemic dummy variable was introduced, taking a value of 1 for the years 2020–2022 and 0 otherwise.

Data for the passenger demand model were retrieved from authoritative institutions deemed highly reliable. More specifically, Eurostat played a pivotal role in data collection, providing variables such as passenger volume, total population, accommodation capacity, educational attainment, as well as financial and economic indicators including the Consumer Price Index (CPI), Purchasing Power Parity (PPP), and jet fuel consumption.

Another key institution in the data collection process was the World Bank, from which real GDP per capita, exchange rates, urban population figures, and international tourist arrivals were extracted.

Our sample consists of airports spanning multiple categories. More specifically, it comprises major European gateways—that is, large hub airports characterized by massive passenger volumes and complex infrastructure. Additionally, the sample includes seasonally and tourism-oriented airports, where traffic spikes exclusively during their peak operational period. Furthermore, our dataset encompasses airports located in isolated regions, such as islands, where air transport serves as an essential medium for rapid and efficient connectivity.

Table 8: List of Sample European Airports ($N = 51$)

#	IATA / ICAO	Airport Name	Country	#	IATA / ICAO	Airport Name	Country
1	LHR / EGLL	London Heathrow	UK	27	BUD / LHBP	Budapest Liszt	Hungary
2	CDG / LFPG	Paris Charles de Gaulle	France	28	NCE / LFMN	Nice Côte d’Azur	France
3	AMS / EHAM	Amsterdam Schiphol	Netherlands	29	DUS / EDDL	Düsseldorf Airport	Germany
4	FRA / EDDF	Frankfurt Airport	Germany	30	BER / EDDB	Berlin Brandenburg	Germany
5	MAD / LEMD	Madrid-Barajas	Spain	31	OPO / LPPR	Porto Airport	Portugal
6	BCN / LEBL	Barcelona–El Prat	Spain	32	VCE / LIPZ	Venice Marco Polo	Italy
7	IST / LTFM	Istanbul Airport	Turkey	33	LIN / LIML	Milan Linate	Italy
8	MUC / EDDM	Munich Airport	Germany	34	GVA / LSGG	Geneva Airport	Switzerland
9	FCO / LIRF	Rome Fiumicino	Italy	35	HAM / EDDH	Hamburg Airport	Germany
10	LGW / EGKK	London Gatwick	UK	36	STR / EDDS	Stuttgart Airport	Germany
11	ORY / LFPO	Paris Orly	France	37	MRS / LFML	Marseille Provence	France
12	DUB / EIDW	Dublin Airport	Ireland	38	LYS / LFL	Lyon–Saint Exupéry	France
13	VIE / LOWW	Vienna International	Austria	39	TLS / LFBO	Toulouse–Blagnac	France
14	ZRH / LSZH	Zurich Airport	Switzerland	40	AGP / LEMG	Málaga–Costa del Sol	Spain
15	LIS / LPPT	Lisbon Humberto Delgado	Portugal	41	ALC / LEAL	Alicante–Elche	Spain
16	CPH / EKCH	Copenhagen Airport	Denmark	42	LPA / GCLP	Gran Canaria	Spain
17	MPX / LIMC	Milan Malpensa	Italy	43	TFS / GCTS	Tenerife South	Spain
18	ATH / LGAV	Athens International	Greece	44	BHX / EGBB	Birmingham Airport	UK
19	BRU / EBBR	Brussels Airport	Belgium	45	EDI / EGPB	Edinburgh Airport	UK
20	OSL / ENGM	Oslo Gardermoen	Norway	46	OTP / LROP	Bucharest Henri Coandă	Romania
21	ARN / ESSA	Stockholm Arlanda	Sweden	47	SOF / LBSF	Sofia Airport	Bulgaria
22	HEL / EFHK	Helsinki–Vantaa	Finland	48	RIX / EVRA	Riga International	Latvia
23	MAN / EGCC	Manchester Airport	UK	49	ZAG / LDZA	Zagreb Airport	Croatia
24	PMI / LEPA	Palma de Mallorca	Spain	50	BEG / LYBE	Belgrade Nikola Tesla	Serbia
25	PRG / LKPR	Prague Václav Havel	Czechia	51	SKG / LGRP	Thessaloniki Macedonia	Greece
26	WAW / EPWA	Warsaw Chopin	Poland				

Note: Includes the top 51 European airports operating over the 2014–2024 period based on Eurostat and Eurocontrol operational data.

The table above presents the selected sample of airports. As observed, major international hubs such as Paris Charles de Gaulle (CDG), Amsterdam Schiphol (AMS), Frankfurt (FRA), Munich (MUC), and London Heathrow (LHR) handle immense year-round passenger volumes, thereby driving intra-European and global connectivity.

Additionally, airports such as Santorini (JTR), Mykonos (JMK), Ibiza (IBZ), and Palma de Mallorca (PMI) are defined by strong summer seasonality, given their locations in premier holiday destinations. Conversely, airports like Chambéry-Aix-les-Bains (CMF) and Calvi (CLY) exhibit pronounced winter seasonality, ensuring that the empirical analysis captures diverse operational profiles across Europe.

Finally, the sample includes island-based airports where air transport provides essential accessibility. Specifically, airports across island regions of Greece, Spain, Cyprus, and Iceland reflect high levels of insularity. Incorporating these locations enables an assessment of how geographical constraints impact operational efficiency.

In the demand model, the sample was examined at the country level, as air passenger demand can be effectively investigated as a macroeconomic question. The table below presents the selected countries.

Table 9: List of Included European Countries ($N = 27$)

Country (1–9)	Country (10–18)	Country (19–27)
Austria	Germany	Netherlands
Belgium	Greece	North Macedonia
Bulgaria	Hungary	Norway
Croatia	Iceland*	Poland
Cyprus*	Ireland*	Portugal
Czechia	Italy	Romania
Denmark	Latvia	Spain
Finland	Luxembourg	Sweden
France	Malta*	Switzerland

Note: Asterisk (*) denotes island or geographically isolated states ($geo = 1$).

Our sample comprises a total of 27 diverse European countries over the period 2018–2024. This timeframe captures a contemporary economic landscape while encompassing the COVID-19 pandemic, thereby enabling an evaluation of passenger demand dynamics before, during, and after the crisis. Specifically, the dataset includes advanced economies such as Germany, France, the Netherlands, Belgium, and Switzerland, representing high-GDP markets characterized by mature travel demand, strong purchasing power, and a high share of business travel.

Furthermore, emerging markets of Central and Eastern Europe—including Poland, Romania, Bulgaria, the Czech Republic, Hungary, Croatia, Latvia, and North Macedonia—were included to capture economic convergence and the rapid growth of air transport

in developing economies.

Additionally, key tourist destinations such as Greece, Spain, Portugal, and Italy provide the necessary context for assessing inbound tourism flows and high seasonal variability.

Finally, to evaluate the structural impact of insularity on passenger demand, island nations such as Cyprus, Iceland, Ireland, and Malta were incorporated into the sample. In these regions, the lack of alternative overland infrastructure renders air transport an essential, near-monopolistic mode of transit, structurally altering demand elasticities.

Overall, covering 27 nations over the 2018–2024 period yields a panel of 189 observations, ensuring adequate degrees of freedom for robust econometric estimation without small-sample bias.

3.2 Inputs and Outputs for DEA Model

The choice of the DEA model according to the literature has the advantage of being able to handle multiple variables, meaning it allows the selection of multiple inputs and outputs, making it a more flexible method than parametric methods like SFA. In this chapter, the inputs and outputs used in the efficiency model will be discussed and presented, as well as the variables chosen in the model that identifies the factors affecting airport efficiency, and the output sources will also be identified.

The selection of inputs and outputs for assessing the technical efficiency of the 51 airports in Europe is based on systematizing international experience. As the literature points out, an airport's operation is divided into the infrastructure of the airside and the terminal side.

Starting with the model's inputs, we can see variables related to both the construction of the airport and its operational activity. More specifically, one of the key inputs mentioned in the literature is the number of runways the airport has. In particular, this determines the airport's maximum theoretical capacity for simultaneous landings and takeoffs.

In the literature, its use as a separate input is now a classic approach, with examples from studies by Gillen & Lall (1997), Sarkis (2000), and Pels et al. (2003). It is noted that, although some researchers use standardized routes to avoid differences in dimensions, the number of air runways is considered the most absolute indicator of airport capacity.

Another input of the model used in this research as well as in the international literature is the number of terminal stations. More specifically, the number of terminal stations represents the main ground infrastructure of airports for serving passengers. The use of the number of independent terminal stations as a physical input is very popular in the international literature, with notable examples being Adler & Berechman (2001) and Pedram & Payan (2015).

The use of the number of terminal stations at an airport is not random, but just as a runway can show the capacity of aircraft an airport can accommodate, the terminal station reflects the capacity of passengers it can serve and host at the airport. More specifically, this indicator very effectively captures the complexity and size of the ground facilities managing passenger flows. So, for all these reasons, the number of terminal stations is considered a key input that has been included in the efficiency assessment model.

In the end, regarding outflows, we will continue with the airport's financial data as a business. More specifically, we will talk about the labor cost that the airport has, meaning the cost that the business needs to manage based on the volume of staff it has. This is a financial variable used in international literature as an input because it reflects human capital and the operating employment expense of the airport, meaning it shows the amount of human support the airport has for managing passenger and aircraft traffic during boarding and disembarking, as well as administrative staff and the organization of the airport's apron. Research that uses this variable for the same reasons includes Fragoudaki & Giokas, 2020; Oum et al., 2008.

Continuing with the outflows, the literature includes economic variables but also variables related to aircraft movement and cargo.

Initially, an important outflow is the movements of aircraft or ATM, as the literature often refers to it. More specifically, international literature points out that total take-offs and landings are considered among the most classic aerial outflows in the literature, with specific notable studies being Gillen & Lall, 1997; Pels et al., 2003.

Two other variables that are very often linked to an airport's outputs are passenger demand and the amount of cargo handled. In the literature, however, they often use another variable called WLU or Workload Units, which combines the number of passengers as well as the volume of cargo handled. As shown in the equation below, WLU is a composite variable produced as follows:

$$WLU = Passengers + 10CargoTones \quad (1)$$

This variable is used because the number of passengers and the cargo are two important outputs, but due to high correlation, this one is used as a combination of the two above.

The last outflow added to the model concerns the financial category of the model. More specifically, we are talking about the airport's aeronautical revenues. This variable helps to be considered as an outflow since it includes both the aeronautical and commercial revenues of the airport, reflecting its commercial success. Notable studies that include this variable as an outflow are Fragoudaki & Giokas, 2020; Yilmaz et al., 2022.

In this research, this outflow is added to the DEA model as it adopts the international

practice of Martín et al. (2009).

Table 10: Identification and Documentation of DEA Model Variables

Variable	Symbol	Unit of Measurement	Literature Documentation	Docu-Data Source
Inputs				
Number of Runways	Runways	Count	Gillen & Lall (1997) [?], Pels et al. (2003) [17], Sarkis (2000) [?]	<i>Airport official information site</i>
Number of Terminals	Terminals	Count	Adler & Berechman (2001) [?], Pedram & Payan (2015) [?]	<i>Airport official information site</i>
Labor Costs	Labor Costs	Euro (€)	Fragoudaki & Giokas (2020) [16], Oum et al. (2008) [?]	<i>Annual Reports</i>
Outputs				
Aircraft Movements	ATM	Number of flights	Gillen & Lall (1997) [?], Pels et al. (2003) [17]	<i>Eurostat</i>
Work Load Units	WLU	WLU units	Fragoudaki & Giokas (2020) [16], Jo & Chang (2023) [?]	<i>Eurostat</i>
Operating Revenues	Revenues	Euro (€)	Fragoudaki & Giokas (2020) [16], Yilmaz et al. (2022) [18]	<i>Annual Reports</i>

Table 7 briefly shows the variables of the inputs and outputs, the names of the variables given to be used in the model's code, their units of measurement, as well as the papers that mention them as outputs and correspondingly as inputs.

Continuing then in this chapter with the data sources and from which organizations the data were found. Initially, starting with the inputs, the data on the number of runways and terminal stations were found on the official pages of the airports. Also, the variable for personnel costs was found from the official financial reports each year for each airport in the sample.

Additionally, for the output variables, the data used were found from various organizations that have freely available databases for the general public. More specifically, the WLU variable was created by applying the above formula, using data from Eurostat for cargo data from 2014 to 2024 annually, and from the same source, data were also found for the other variable in the equation, that is, the number of passengers annually at the country level.

Moving on to the factors that affect airport efficiency. In the literature, there are factors mentioned from economic, geographical, socio-demographic categories, as well as crises, depending on the study. In the present study, as the literature also points out, in

the second stage of finding efficiency, such categories of variables are included, as well as crisis variables, like we mentioned earlier, a variable referring to the COVID-19 pandemic.

Starting with the economic variables, the model includes the per capita real GDP variable, which according to the literature represents the economic level of the airport's country. In general, it is positively associated with efficiency according to Kaya et al., 2022. However, Fragoudaki & Giokas (2016) mention that in peripheral tourist markets this variable may not be statistically significant, as efficiency is determined by other factors.

Another variable that is added is the country's educational level. According to Kluge et al. (2017), this variable contributes to increased efficiency because the higher the country's educational level, the more people tend to travel by air.

At the same time, the literature talks about the variable of total population. More specifically, it is used as an indicator of the size of the local market. While theoretically a larger population increases demand, its effect on efficiency can be negative due to the phenomenon of infrastructure congestion, as argued by the theory of Tsekeris (2011) and Fernandes & Pacheco (2002).

Also, alongside the population variable, the literature uses the variable of tourist arrivals, which reflects the total tourist demand in the area. According to the literature, the results of this variable are expected to be positive, as Fragoudaki & Giokas, 2016; Yilmaz et al., 2022 argue that airport efficiency is tourism-driven.

Also, the hotel capacity of countries plays a crucial role in airport efficiency since it reflects the capacity of local hospitality infrastructure. As Fragoudaki & Giokas, 2016; Dieu Huynh et al., 2025 argue, it acts as a supporting factor in tourism demand and is expected to have a positive effect.

A key factor that makes air transport essential is traveling from a country to a solid geographic location that is an island. So, islandness plays a crucial role in efficiency according to the literature, and more specifically, Tsekeris (2011) and Fragoudaki & Giokas (2016) note that a positive effect on efficiency is expected, as island airports serve as monopoly entry points for international tourism, showing high operational density.

Finally, as we said at the beginning, crises play a big role in an airport's efficiency. The COVID-19 crisis was a harsh negative factor on airport efficiency, and according to Gao, 2022; Dieu Huynh et al., 2025, it represented a phenomenon of complete stagnation. In the table below, the variables that were discussed and that will serve as independent variables in our model are summarized.

Table 11: Identification and Documentation of Exogenous Variables (Tobit Model)

Variable	Symbol	Expected Sign	Literature Reference	Reference	Data Source
GDP per Capita	log(GDPpc)	+ / +/−	Kaya et al. (2022) [7], Fragoudaki & Giokas (2016) [14]		World Bank
Tourist Arrivals	log(Tourists)	+	Fragoudaki & Giokas (2016) [14], Yilmaz et al. (2022) [18]		World Bank
Accommodation Capacity	log(Accom)	+	Fragoudaki & Giokas (2016) [14], Huynh et al. (2025) [19]		Eurostat
Total Population	log(Population)	− / +	Tsekeris (2011) [10], Yilmaz et al. (2022) [18]		Eurostat
Insularity (Dummy)	geo	+	Tsekeris (2011) [10], Fragoudaki & Giokas (2016) [14]		Own Construction / Geographical Data
Educational Level	education	+	Kluge et al. (2017) [?]		Eurostat
Pandemic (Dummy)	Covid_19	−	Gao (2022) [?], Huynh et al. (2025) [19]		Own Construction / Time Period Dummy

The above variables came from organizations that provide databases for free as well as some others, and dummy variables were created without a source. More specifically, the variables for real GDP per capita and tourist arrivals came from the World Bank. The variables related to the country’s hotel capacity, total population, and education level came from Eurostat. Finally, the island dummy variables were indicated to take the value 1 if the country or location is on an island and 0 otherwise, and for the pandemic variable, it was indicated as 1 if the year falls during the COVID pandemic and 0 otherwise.

3.3 Passenger Demand variables

This chapter discusses the variables selected for the passenger demand model, outlining the theoretical rationale and the academic literature from which they were drawn. The selection of these model variables reflects the socioeconomic and country-level characteristics of the analyzed sample.

Table 12: Identification, Expected Signs, and Literature Documentation of Demand Variables

Variable	Symbol	Expected Sign	Literature Reference	Data Source
Passengers (Dep)	<i>Passengers</i>	<i>Dependent</i>	Valdes (2015) [13], Huynh et al. (2025) [19]	Eurostat
Real GDP per Capita	<i>GDPpc</i>	+	Valdes (2015) [13], Kluge et al. (2017) [?]	World Bank
Total Population	<i>POP</i>	+	Dayioglu & Alnipak (2023), Akinyemi (2019) [?]	Eurostat
Urban Population (%)	<i>URB</i>	+	Kluge et al. (2017) [?], Iyer & Thomas (2021)	World Bank
Tourist Arrivals	<i>TOUR</i>	+	Dieu Huynh et al. (2025) [19]	World Bank
Accommodation Capacity	<i>Accom</i>	+	Dieu Huynh et al. (2025) [19], Pels et al. (2003) [17]	Eurostat
Consumer Price Index	<i>CPI</i>	−	Dayioglu & Alnipak (2023), Avcı & Avcı (2025) [20]	Eurostat
Exchange Rate	<i>EX</i>	+ / −	Avcı & Avcı (2025) [20], Gelirli (2022)	World Bank
Fuel Consumption (Tons)	<i>FUEL</i>	− / +	Abate (2016), Simić & Babić (2015) [?]	Eurostat
Insularity (Dummy)	<i>geo</i>	+	Kluge et al. (2017) [?], Grosche et al. (2007) [9]	Own Construction / GIS Data
Education Level	<i>educ</i>	+	Kluge et al. (2017) [?]	Eurostat
Purchasing Power Parity	<i>PPP</i>	+	Oum et al. (2008) [?]	Eurostat
Pandemic (Dummy)	<i>Covid_19</i>	−	Suau-Sanchez et al. (2020), Gao (2022)	Own Construction / Time Period Dummy

The table above summarizes the model’s variables alongside the respective studies in which they have been utilized. Regarding passenger demand, the dependent variable is defined as the annual number of passengers per country, aligned with Valdes (2015), Dieu Huynh et al. (2025), Dayioglu & Alnipak (2023), and Avcı & Avcı (2025), as passenger traffic represents the most direct and reliable proxy for total manifested demand for air transport.

The independent variables in the model are multifaceted, spanning several categories. More specifically, Valdes (2015), Kluge et al. (2017), and Dayioglu & Alnipak (2023) incorporate Real GDP per capita as the primary driver of passenger demand, under the premise that an increase in a country’s economic standing and living standards boosts travel “consumption” for both leisure and business purposes.

Furthermore, to capture true purchasing power, Dayioglu & Alnipak (2023) introduce Purchasing Power Parity (PPP) into their framework as an indicator of consumer purchas-

ing capacity beyond nominal income levels, which positively influences travel demand. The same authors, along with Valdes (2015), also include Foreign Direct Investment (FDI) as a proxy for business activity that directly stimulates business-related air travel.

Regarding demographic and urbanization factors, total population is employed by Dayioglu & Alnepak (2023) and Abate (2016) to represent potential market size, expanding the base of potential trips. Additionally, Kluge et al. (2017) and Dayioglu & Alnepak (2023) include urban population, arguing that high urbanization implies a concentration of economic activity, which in turn increases travel frequency toward urban centers and regions of origin.

In terms of tourism drivers and hospitality infrastructure, tourist arrivals and regional accommodation capacity are drawn from Dieu Huynh et al. (2025), Pels et al. (2003), and Dayioglu & Alnepak (2023). According to these studies, accommodation capacity acts as a key pull factor for passenger demand, while the sufficiency of tourist infrastructure directly boosts air arrivals.

Monetary and price indicators, specifically the Consumer Price Index (CPI) and exchange rates, are utilized by Valdes (2015), Dayioglu & Alnepak (2023), and Avcı & Avcı (2025). The CPI reflects living cost inflation that squeezes disposable income available for travel, whereas the exchange rate directly impacts the cost of traveling to foreign currency destinations.

A critical role in the literature is played by aviation fuel (jet kerosene) consumption. Valdes (2015) and Abate (2016) employ this variable as a cost proxy for airlines; higher fuel expenses inflate airfares, thereby expecting a reduction in demand due to ticket price increases.

Geographic characteristics significantly influence passenger movement. Accordingly, Kluge et al. (2017) and Grosche et al. (2007) incorporate geographic location into their models to test whether demand increases at island-located airports. Consequently, the geographic variable (geo) is constructed as a dummy variable that plays an essential role in the model's regression.

Educational attainment also constitutes an important factor in travel behavior, as academic pursuits and educational trips make air travel essential. Kluge et al. (2017) argue that higher levels of education are associated with employment in knowledge-intensive sectors and greater individual mobility, thereby enhancing both professional and personal travel.

Finally, because the dataset spans the COVID-19 pandemic, including a pandemic dummy variable becomes necessary to evaluate demand behavior during severe crisis periods, particularly given the lockdowns associated with COVID-1

3.4 Data manipulation

This chapter will describe the methods used to prepare the data for analysis. More specifically, due to the large volume of data from different sources, there may be observations that are empty—that is, with no values recorded.

To begin with, the selected datasets included less developed countries whose data had not been recorded in this format; therefore, to improve the model’s fit, they were removed. However, even after the deletion, there were still observations that were NaN; to fill them in, another method called linear interpolation was used, which essentially accurately estimates an unknown value between two known values, assuming that the change occurs at a constant, uniform rate. The equation for linear interpolation is as follows:

$$y = y_1 + (x - x_1) \frac{y_2 - y_1}{x_2 - x_1} \quad (2)$$

The equation finds an unknown intermediate value (y) by assuming that two known points are connected by a straight line. First, it calculates the constant rate of change (slope) between the two known endpoints using the fraction. It then multiplies this slope by the distance of x from the starting point and adds it to the initial height (y_1).

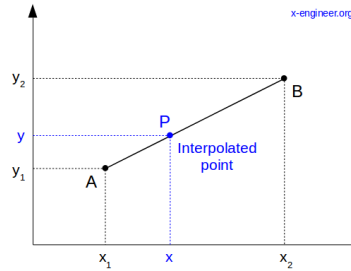


Figure 1: Linear interpolation

3.5 Descriptive statistics for the inputs and outputs of the DEA model variables and Exogenous Determinants of efficiency.

This subsection presents the descriptive statistics for the input and output variables of the DEA model, as well as the exogenous variables affecting efficiency. The table below outlines the descriptive statistics of the variables included in the model.

From the table, several conclusions can be drawn regarding our sample, as the variables of the DEA model exhibit pronounced asymmetry and structural heterogeneity across the examined European airports. Across all evaluated variables, a substantial divergence is observed between the mean and the median, alongside exceptionally high standard deviations.

Furthermore, right-skewness is evident, driven by the coexistence of massive European

Table 13: Descriptive Statistics of Airport Efficiency Metrics and Input/Output Variables

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
<i>Inputs & Outputs</i>								
WLU (Output)	650	13,858,367	19,842,544	2,525	1,607,955	4,736,881	18,349,760	97,498,510
Movements	650	2,130,556	5,618,290	108	39,268	132,190	477,030	31,617,148
Labor Cost	650	58,181,661	92,207,249	1,000	9,510,500	21,399,500	68,072,500	662,817,000
Operating Cost	650	109,726,929	179,350,917	2,000	13,446,750	36,997,500	122,238,500	1,225,581,000
Gates	650	28.53	37.10	1.00	5.00	14.00	36.75	212.00
Stands	650	50.81	59.98	1.00	11.00	28.00	66.00	398.00
Runways	650	1.94	1.08	1.00	1.00	2.00	2.00	7.00
Terminal (sqm)	650	116,367	165,854	1,200	17,219	46,250	135,150	1,200,000

mega-hubs alongside smaller regional, seasonal, and tourism-oriented airports within the same dataset.

Starting with the outputs, the Work Load Units (WLU) variable serves as the primary composite output, integrating annual passenger traffic and cargo tonnage. As shown in the table, the mean annual value for WLU stands at 13.86 million units, whereas the median is merely 4.74 million units. Values range from a minimum of 2,525 WLU to a maximum of 97.5 million WLU among major European hubs.

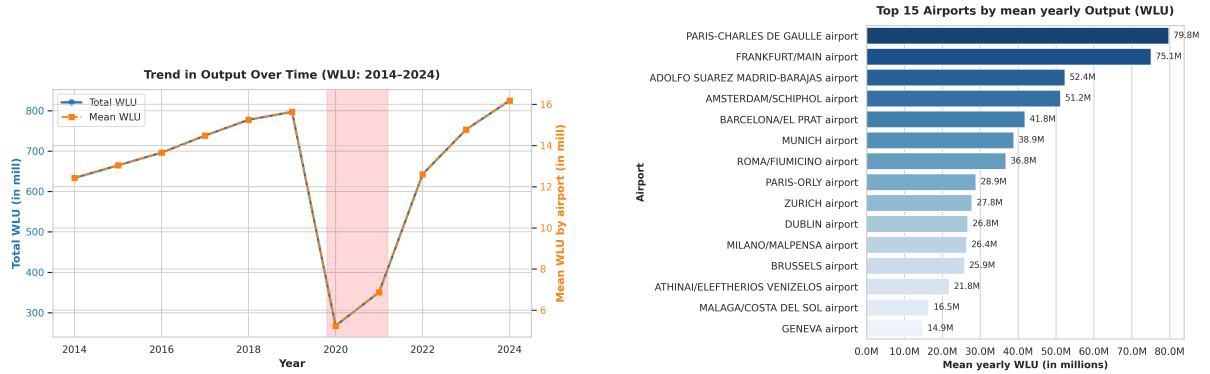
Regarding Aircraft Movements (ATM), which capture flight operations (takeoffs and landings), the mean is 2.13 million movements, with a median of 132,190 movements and a peak value reaching 31.6 million movements.

Finally, concerning financial outputs, Operating Revenues reflect the airport’s overall performance across aeronautical and non-aeronautical activities. Mean annual operating revenues stand at €109.7 million, with a median of €37 million. The maximum recorded figure reaches €1.225 billion, confirming the immense financial power of European mega-hubs.

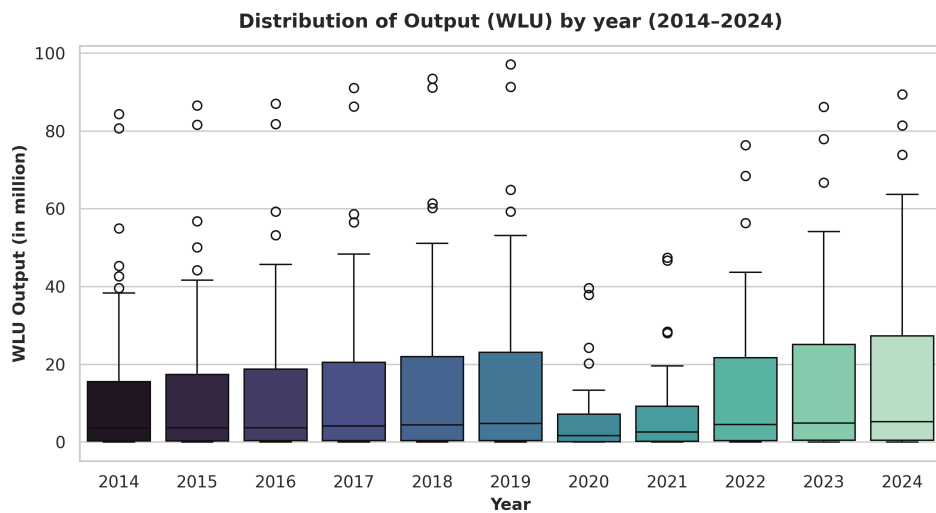
Turning to the inputs, these variables characterize both the physical infrastructure and financial profile of the airports. First, the number of runways averages 1.94 runways, with a median of 2, a standard deviation of 1.08, and a range spanning from a minimum of 1 runway to a maximum of 7 for major international airports. Conceptually, this represents the primary airside infrastructure input determining maximum simultaneous landing and takeoff capacity. The median value of 2 runways confirms that the majority of European airports operate with 1 to 2 runways. Because runways represent an indivisible capital input that cannot be scaled down in the short run, maintaining them during periods of declining demand (such as the pandemic) generates structural excess capacity.

With respect to Labor Costs, which reflect an airport’s workforce expenditures, the annual average is €58.18 million, while the median is €21.40 million—more than 2.7 times lower. The minimum value fell to €1,000 during the periods of total lockdowns during the pandemic, whereas the maximum surged to €662.82 million, with a standard deviation of €92.21 million. The vast gap between the mean and median underscores the heavy concentration of personnel at major hubs. Concurrently, the longitudinal trend

demonstrates that labor expenses decreased during the pandemic (2020–2021) due to layoffs and furloughs; however, this reduction was far milder than the steep collapse in operating revenues, highlighting the relative rigidity of human resources in infrastructure sectors.



(a) Longitudinal evolution of total and average WLU (2014–2024). (b) Top 15 airports ranked by mean annual Output (WLU).



(c) Annual distribution and dispersion of Output (WLU).

Figure 2: Statistical and time-series analysis of the output variable (WLU Output) across the sample of European airports.

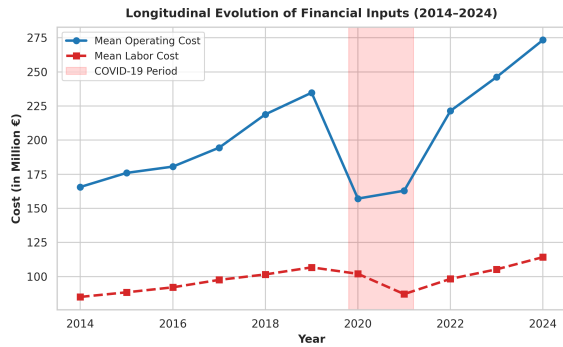
From the graphs above, we can draw several conclusions about our sample. To begin with, in Graph 9a, we can observe a trend over time and the impact of the pandemic on the composite variable WLU. More specifically, in the period before the pandemic, we observe an upward trend in passenger and freight traffic—rising from approximately 630 million in 2014 to nearly 800 million in 2019.

During the pandemic, we observed a sharp decline, bringing the average annual WLU per airport below 5.5 million. Following the pandemic, we are seeing a recovery in the sector as the economy strives to return to its normal level, entering a recovery phase that began gradually in 2021, and by 2024, total output had surpassed pre-pandemic levels,

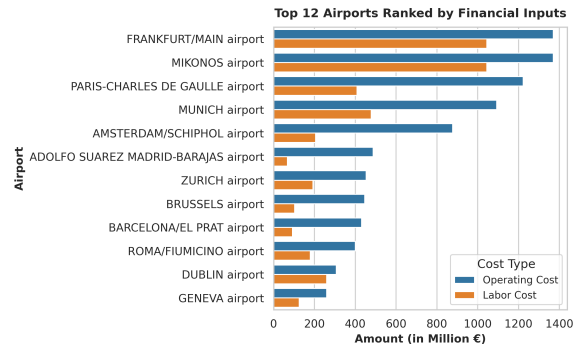
reaching 820 million WLU.

In Figure 9b, we can see the ranking of airports based on the average value of the WLU variable. We can say that the airports are characterized by pronounced structural asymmetry, as the top of the list is dominated by major European hubs, such as Paris-Charles de Gaulle (79.8M WLU) and Frankfurt/Main (75.1M WLU), followed by the airports in Madrid (52.4M WLU) and Amsterdam (51.2M WLU). Athens Airport ranks 13th with an average annual volume of 21.8 million WLU.

Finally, we have Figure 9c, a boxplot that helps us understand how the sample's variance is distributed. More specifically, The annual boxplots confirm the persistent positive skewness of the data. In all years, the median remains noticeably lower than the arithmetic mean, while a significant number of positive outliers are observed. These values represent the mega-hubs in the sample, which essentially define the production frontier in the nonparametric DEA model.



(a) Longitudinal evolution of mean operating and labor costs (2014–2024).



(b) Top 12 airports ranked by mean annual financial inputs.

Figure 3: Statistical and time-series analysis of the input variables across the sample of European airports.

Figure 10a shows the average trend of the financial variables for the airports in the study sample. More specifically, during the pre-pandemic period—that is, from 2014 to 2019—we observe a steady upward trend in both categories of expenses; specifically, operating expenses rose from 165 million in 2014 to 235 million in 2019. Labor costs also followed a similar upward trend, peaking at 107 million in the year

During the pandemic—that is, in 2020, 2021, and 2022—as shown by the shaded area in the graph—there was a sharp drop in operating expenses, specifically a decrease of 155 million euros; labor costs also decreased, but not to the same extent as operating expenses. Specifically, these two trends can be attributed to the significant reduction in flights, which led to massive layoffs of employees in the industry. More generally, one could characterize the situation as a crisis affecting both the industry and its employees.

After the crisis—that is, in 2023 and 2024—we see a rapid return to 2019 levels. In

2024, there is a historic increase to 270 million euros in operating expenses, as well as inflationary pressures on energy and maintenance costs. This is due to the full recovery of passenger traffic and the increase in the workforce in the sector.

Figure 10b shows a graph of the top 12 airports in terms of their financial performance. At first glance, there appears to be a strong concentration of expenses at the largest airports in Europe, specifically the airports in Frankfurt, Paris, Munich, and Amsterdam. The study also mentions specific tourism-related cases, such as Mykonos Airport in Greece. In conclusion, at all airports on the list, operating expenses are significantly higher than labor costs. This demonstrates that the aviation industry is extremely capital-intensive, requiring enormous expenses for the maintenance and operation of terminals as well as technological equipment relative to the direct cost of employee wages.

Table 14: Descriptive statistics of Tobit model environmental determinants.

Variable	N	Mean	Std. Dev.	Min	Median	Max
Real GDP per Capita (€)	357	55,848.00	20,105.58	23,997.69	52,466.73	138,522.50
Tourist Arrivals (M)	357	29.15	24.83	0.71	23.10	77.67
Total Population (M)	357	33.31	26.73	0.35	17.48	83.46
Urban Population (%)	357	78.83	5.60	63.08	78.78	95.64
Education Level (%)	357	37.85	7.70	19.40	39.70	56.60
Consumer Price Index (CPI)	357	2.85	2.74	-1.30	2.30	13.00
Exchange Rate	357	95.16	8.88	73.67	95.57	140.46
Purchasing Power Parity	357	6.93	33.57	0.83	1.06	257.89
Accommodation Capacity (K)	357	1,102.41	775.68	32.22	906.87	2,283.55
Geographic Dummy (Island)	357	0.039	0.194	0.000	0.000	1.000
COVID-19 Dummy	357	0.429	0.496	0.000	0.000	1.000

Regarding the exogenous factors affecting the efficiency of the sample airports, we proceed by incorporating variables across multiple categories into our model. The table below presents the descriptive statistics of these variables, specifically covering categories such as economic, demographic, macroeconomic, and spatial conditions that influence the operational efficiency of European airports.

Starting with the economic and demographic indicators, Real GDP per Capita exhibits a mean value of €55,848.00, with a high standard deviation of €20,105.58 and a wide variation range between €23,997.69 and €138,522.50. This disparity reflects the diverse economic levels across the countries included in our airport sample. Total Population averages 33.31 million inhabitants (ranging from 0.35 to 83.46 million), while Urban Population (%) averages 78.83% with a low standard deviation of 5.60%, confirming that the majority of the analyzed airports serve highly urbanized regions. Furthermore, Education Level records an average of 37.85%, reflecting variations in regional human capital.

Moving on to tourism indicators, International Tourist Arrivals average 29.15 million per country, ranging from 0.71 to 77.67 million. Correspondingly, Tourist Accommoda-

tion Capacity averages 1,102.41 thousand bed places with a median of 906.87 thousand, capturing the physical hospitality infrastructure supporting passenger flows.

Regarding macroeconomic determinants, the Consumer Price Index (CPI) averages 2.85%, reaching a peak of 13.00%, which captures the sharp inflationary pressures experienced during the post-pandemic recovery phase. Purchasing Power Parity (PPP) exhibits a mean value of 6.93 with positive skewness, driven by specific European regions characterized by a high cost of living. Finally, the Exchange Rate index maintains relative stability around a mean value of 95.16 units, with a median of 95.57.

Below is the graphical representation of the model variables for the exogenous factors affecting airport efficiency. These boxplots summarize the distribution of the dataset across the analyzed variables.

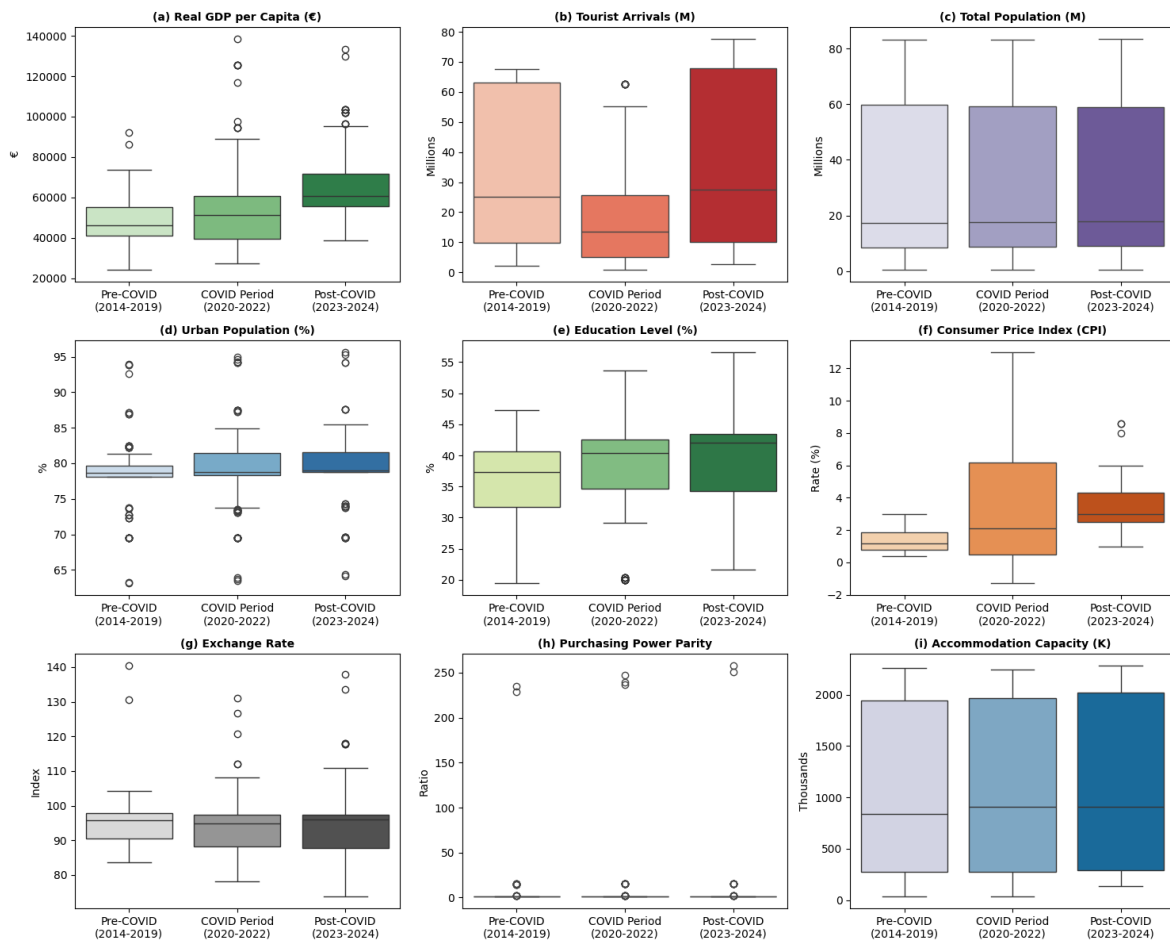


Figure 4: Distributional shifts across environmental and macroeconomic determinants in the Tobit model over Pre-COVID (2014–2019), COVID-19 (2020–2022), and Post-COVID (2023–2024) periods.

Figure 3 illustrates the longitudinal evolution of the environmental and macroeconomic variables in the Tobit framework across the three study periods: Pre-COVID, the COVID Period, and Post-COVID. It is clearly observed that structural demographic and physical indicators maintain exceptional stability over time: Total Population (c), Urban

Population (d), and Accommodation Capacity (i) display nearly unchanged distributions, confirming that core infrastructure and demographic scales remained unaffected by the pandemic shock. Meanwhile, Real GDP per Capita (a) and Education Level (e) record a steady upward trend in their median values, although GDP dispersion widens noticeably in the Post-COVID period due to the uneven economic recovery among European states.

Conversely, variables directly linked to travel demand and macroeconomic stability experienced severe shocks. International Tourist Arrivals (b) suffered a steep decline during the pandemic period (2020–2022), before fully rebounding in 2023–2024 to reach pre-pandemic peak levels of nearly 78 million. Simultaneously, the Consumer Price Index shifted from absolute monetary stability in the Pre-COVID period (1.2%) to a phase of intense volatility during the pandemic, culminating in a permanently higher median (3.5%) in the Post-COVID period accompanied by extreme inflationary outliers. Finally, the Exchange Rate and Purchasing Power Parity maintain stable medians with pronounced positive skewness, driven by specific high-cost economies.

3.5.1 Descriptive statistics for passenger demand model

In this chapter, we will discuss and visualize the descriptive statistics for the variables that, according to the literature, are factors influencing passenger traffic in Europe. Passenger traffic is an integral part of airport efficiency, as we saw earlier, the arrival of passengers at airports tests their operational capabilities, and in this way, the airport can improve its performance and the way it handles the continuous increase in the number of travelers. This dataset includes data at the country level as well as time series from 2018 through 2024. More specifically, the table below shows these statistics.

Table 15: Summary Statistics of Air Passenger Demand Model Variables.

Variable	N	Mean	Std. Dev.	Min	Median	Max
Passengers (M)	189	41.20	54.97	0.71	19.11	259.74
Education Level (%)	189	36.63	9.39	17.80	37.90	56.60
Real GDP per Capita (€)	189	58,715.60	26,122.95	18,460.46	55,761.38	155,941.29
Urban Population	189	42,170.96	215,212.09	52.16	78.39	1,141,402.00
Tourist Arrivals (M)	189	12.33	17.03	0.12	5.49	77.67
Total Population (M)	189	16.72	21.85	0.35	8.96	83.46
Consumer Price Index (CPI)	189	3.66	3.58	-1.30	2.60	17.20
Exchange Rate	189	98.31	10.56	73.67	97.14	140.46
Geographic Dummy (Island)	189	0.148	0.356	0.000	0.000	1.000
COVID-19 Dummy	189	0.429	0.496	0.000	0.000	1.000
Purchasing Power Parity	189	21.70	61.56	0.65	1.21	270.11
Tourism Employment (%)	189	15.40	20.31	0.50	7.50	90.30
Accommodation Capacity (K)	189	472.67	619.08	14.40	233.55	2,283.55

To begin with, we will discuss the model’s dependent variable, which is the “passengers” variable. More specifically, at the country level stands at 41.2 million. However,

we observe a wide variation within the sample: in the smaller countries, the figure ranges from 0.71 million, while in Europe’s major aviation markets, it ranges from 259.74 million—the maximum value—with a median of 19.11 million, which confirms that most European countries fall squarely within the mid-range of national demand.

With regard to economic and demographic indicators, we also observe clear trends. More specifically, the average national GDP per capita stands at 58,715.60, reflecting the purchasing power of domestic consumers for air travel. As for the national population of Europe, it stands at 16.72 million, covering a wide range from small countries (0.35 million) to countries with large populations (83.46 million). Furthermore, with regard to the education variable, it appears that in the countries in the sample where 36.63% of the population holds a higher education degree.

Finally, regarding macroeconomic and geographic variables. More specifically, the CPI index—with an average of 3.66% and a peak of 17.2%—reflects the waves of inflation that affected European economies.

4 Methodology of Analysis

This chapter will present the methodology used in the research to obtain results. More specifically, it will analyze the methodology for handling data and preparing databases for analysis, and will present methods for filling in gaps as well as methods that helped reduce NAN gaps.

We will also present the DEA method—that is, the method used to determine efficiency levels—as well as the “parameters” that were set so that we could derive our results. We will also relate this to the literature—specifically, how other researchers have used the same method and why.

It will also analyze the part of the Tobit model—that is, the model that identified the factors affecting airport efficiency. The model’s equation and its variables will be presented, along with the rationale for its selection.

Finally, the demand section will be analyzed; that is, the selected models and their variables will be examined—specifically, the equation that identifies the factors influencing passenger demand will be analyzed.

4.1 DEA model methodology

To evaluate model efficiency, this thesis employs the non-parametric Data Envelopment Analysis (DEA) method. The DEA methodology enables the simultaneous management of multiple inputs and outputs without requiring the a priori specification of a particular functional form for the production function (Ripoll-Zarraga et al., 2020).

In the aviation sector, because core airport infrastructure represents fixed physical

capital that is inflexible in the short run, an output-oriented framework is adopted (Shephard, 1970). Under this approach, management seeks to maximize the produced aviation volume given the existing level of infrastructure.

We consider a sample of N airports. Each airport i uses a vector of M inputs $x = (x_1, x_2, \dots, x_M) \in \mathbb{R}_+^M$ to produce a vector of K outputs $y = (y_1, y_2, \dots, y_K) \in \mathbb{R}_+^K$. The production possibility set T is defined as:

$$T = \{(x, y) : \text{the input vector } x \text{ can produce the output vector } y\}$$

The corresponding output set is defined as $P(x) = \{y \in \mathbb{R}_+^K : (x, y) \in T\}$ based on Shephard's direct output distance function $D_o(x, y) = \inf\{\delta > 0 : (y/\delta) \in P(x)\}$ the technical efficiency index $\hat{TE}(x, y) \equiv \hat{\theta}(x, y)$ for the evaluated airport o ($o \in \{1, \dots, N\}$) under the assumption of Variable Returns to Scale (VRS) (Banker, Charnes, & Cooper, 1984) is formulated as follows:

$$\begin{aligned} & \max_{\theta, \gamma} \quad \hat{\theta}_o \\ & \text{subject to} \quad \sum_{i=1}^N \gamma_i y_{ki} \geq \hat{\theta}_o y_{ko}, \quad k = 1, \dots, K \\ & \quad \quad \quad \sum_{i=1}^N \gamma_i x_{mi} \leq x_{mo}, \quad m = 1, \dots, M \\ & \quad \quad \quad \sum_{i=1}^N \gamma_i = 1 \\ & \quad \quad \quad \gamma_i \geq 0, \quad i = 1, \dots, N \end{aligned} \tag{3}$$

where $\hat{\theta}_o \geq 1$ represents the proportional output expansion factor and γ_i denotes the intensity variables forming the relative efficiency frontier. The technical efficiency score is defined as $TE_o = \frac{1}{\hat{\theta}_o} \in (0, 1]$, where a value of 1 indicates that the airport lies directly on the fully efficient frontier.

The impact of exogenous economic, geographical, socio-demographic factors, and health crises on the operational efficiency of airports extends beyond the boundaries of internal production functions. These factors act as drivers for improved airport management and dictate how physical infrastructure, workforce, and capacity allocation are

managed. In the present study, the model employed by Fragoudaki & Giokas (2016), Ngo & Tsui (2020), and Yilmaz et al. (2022) is adopted; specifically, a Tobit model is utilized.

In the second stage, the pure technical efficiency scores derived from the first-stage output-oriented DEA VRS model—ranging between 0 and 1—are used as the dependent variable in a censored regression econometric framework. The econometric equation is specified as follows:

$$\begin{aligned} \text{VRS Efficiency}_{it} = & \beta_0 + \beta_1 \ln(\text{GDPpc}_{it}) + \beta_2 \ln(\text{TOUR}_{it}) + \beta_3 \ln(\text{Accom}_{it}) \\ & - \beta_4 \ln(\text{POP}_{it}) - \beta_5 \text{Covid_19}_{it} + \beta_6 \text{geo}_i + \beta_7 \text{educ}_{it} + u_i + \varepsilon_{it} \end{aligned} \quad (4)$$

where β_0 represents the intercept term, u_i denotes the unobserved airport-specific random effect, and $\varepsilon_{i,t}$ is the stochastic error term. The vector of explanatory variables includes Real GDP per capita (Kaya et al., 2022; Fragoudaki & Giokas, 2016), Tourist Arrivals (Fragoudaki & Giokas, 2016; Yilmaz et al., 2022), Accommodation Capacity (Fragoudaki & Giokas, 2016; Dieu Huynh et al., 2025), Total Population of the catchment area (Tsekeris, 2011; Yilmaz et al., 2022), a dummy variable for the pandemic (Covid-19) (Gao, 2022; Dieu Huynh et al., 2025), Insularity/Geography (Tsekeris, 2011; Fragoudaki & Giokas, 2016), and the region’s tertiary Education level (Kluge et al., 2017).

Selecting the appropriate econometric estimator in the second-stage DEA evaluation is a subject of extensive debate in the benchmarking literature (Simar & Wilson, 2007; Ngo & Tsui, 2020). Conventional Ordinary Least Squares (OLS) regressions are mathematically unsuitable for modeling DEA scores, as the dependent variable is bounded and upper-censored at unity (1.000) (Tobin, 1958; Fragoudaki & Giokas, 2016). Applying OLS to bounded dependent variables violates standard normality assumptions, leading to biased and inconsistent estimates (Tobin, 1958; Gutierrez & Lozano, 2016). To accommodate the censored structure of the efficiency scores, a Random-Effects Tobit regression model is implemented (Tobin, 1958; Merkert & Hensher, 2011; Fragoudaki & Giokas, 2016). The Random-Effects specification is particularly well-suited for panel data analysis, as it accounts for unobserved heterogeneity across airports while retaining time-invariant structural variables—such as Insularity—which would otherwise be eliminated under a Fixed Effects transformation (Fragoudaki & Giokas, 2016; Ngo & Tsui, 2020).

Alternative semi-parametric approaches, such as the truncated Bootstrapping regression proposed by Simar and Wilson (2007), were developed to address potential serial correlation and separability issues in two-stage DEA models (Simar & Wilson, 2007; Daraio et al., 2018; Emrouznejad & Yang, 2018). Nevertheless, the Random-Effects Tobit model remains widely established, well-documented, and extensively applied in empirical airport efficiency research (Fragoudaki & Giokas, 2016, 2020; Ngo & Tsui, 2020). Consequently, the Random-Effects Tobit estimator provides a robust, statistically valid, and

highly interpretable econometric framework for identifying how macroeconomic trends, tourism infrastructure, geography, and crises shape the operational efficiency of European airports.

4.2 Passenger Demand Models Methodology

The analysis of passenger demand for air travel constitutes a complex macroeconomic question in the air transport economics literature, influenced by a broad spectrum of economic, demographic, tourism, and geographical determinants. To estimate demand, a panel data econometric approach is adopted. The use of panel data offers significant advantages over simple cross-sectional or time-series data, as it allows for controlling unobserved heterogeneity across countries and increases the degrees of freedom of the estimation.

The dependent variable used is the natural logarithm of the total annual number of passengers per country ($\ln\text{Passengers}$), which serves as the most direct and reliable proxy for revealed air transport demand.

In the literature, total population, tourist arrivals, and accommodation capacity are recognized as fundamental drivers of passenger demand. However, the simultaneous inclusion of these three variables in the same econometric specification induces a severe multicollinearity problem due to the extremely high correlation among them. The presence of multicollinearity inflates standard errors and renders the estimated coefficients statistically insignificant.

To resolve this methodological issue, we follow the approach of Pels et al. (2003) and Ngo & Tsui (2020), who introduce three distinct demand specifications, where each primary variable is evaluated individually while holding the remaining vector of macroeconomic and control variables constant. The specified models are: first, the population model, which examines the impact of total population on the magnitude of passenger demand as an indicator of domestic market size.

The second model incorporates the tourist arrivals variable to capture inbound tourism demand, which represents a primary driver of air travel choice. Finally, the third model utilizes accommodation capacity to evaluate local tourism infrastructure as a factor attracting passenger flows. The estimated models are specified as follows:

$$\begin{aligned}\ln(\text{Passengers}_{it}) = & \beta_0 + \beta_1 \ln(\text{Population}_{it}) + \beta_2 \ln(\text{GDPpc}_{it}) + \beta_3 \ln(\text{UrbanPop}_{it}) \\ & + \beta_4 \text{Education}_{it} + \beta_5 \text{CPI}_{it} + \beta_6 \text{ExchangeRate}_{it} + \beta_7 \text{JetFuel}_{it} \\ & - \beta_8 \text{Covid19}_{it} + \beta_9 \text{PPP}_{it} + \beta_{10} \text{Geo}_i + u_i + \varepsilon_{it}\end{aligned}\quad (5)$$

$$\begin{aligned}\ln(\text{Passengers}_{it}) = & \beta_0 + \beta_1 \ln(\text{TouristArrivals}_{it}) + \beta_2 \ln(\text{GDPpc}_{it}) + \beta_3 \ln(\text{UrbanPop}_{it}) \\ & + \beta_4 \text{Education}_{it} + \beta_5 \text{CPI}_{it} + \beta_6 \text{ExchangeRate}_{it} + \beta_7 \text{JetFuel}_{it} \\ & - \beta_8 \text{Covid19}_{it} + \beta_9 \text{PPP}_{it} + \beta_{10} \text{Geo}_i + u_i + \varepsilon_{it}\end{aligned}\quad (6)$$

$$\begin{aligned}\ln(\text{Passengers}_{it}) = & \beta_0 + \beta_1 \ln(\text{Accommodation}_{it}) + \beta_2 \ln(\text{GDPpc}_{it}) + \beta_3 \ln(\text{UrbanPop}_{it}) \\ & + \beta_4 \text{Education}_{it} + \beta_5 \text{CPI}_{it} + \beta_6 \text{ExchangeRate}_{it} + \beta_7 \text{JetFuel}_{it} \\ & - \beta_8 \text{Covid19}_{it} + \beta_9 \text{PPP}_{it} + \beta_{10} \text{Geo}_i + u_i + \varepsilon_{it}\end{aligned}\quad (7)$$

In this specification, the subscript i represents the country and t denotes the year under examination (2018–2024). The vector of explanatory variables includes the logarithm of Real GDP per capita (GDP_{pc}), reflecting consumer income and purchasing power, alongside the percentage of urban population ($\text{UrbanPop}\%$), capturing demand concentration in metropolitan areas. Additionally, the model incorporates tertiary education attainment ($\text{Education}\%$) as a proxy for travel propensity, the Consumer Price Index (CPI) to account for inflationary pressures, the Exchange Rate, and Jet Fuel consumption, which directly correlates with flight operations. Furthermore, Purchasing Power Parity (PPP) is included to ensure cross-country economic comparability, a dummy variable for insularity/geographic isolation (Geo , taking the value 1 for island or geographically isolated states and 0 otherwise), and a pandemic dummy variable (Covid19 , taking the value 1 for the years 2020–2022 and 0 otherwise). Finally, the model is completed by the unobserved country-specific effect (u_i) and the stochastic error term (ε_{it}).

Because the analysis relies on panel data, applying least squares estimators requires selecting between a Fixed Effects and a Random Effects specification. To guide this choice, Hausman (1978) introduced the Hausman test to evaluate the suitability of random versus fixed effects. Specifically, this statistical test evaluates a null hypothesis (H_0) stating that the individual unobserved effects are uncorrelated with the explanatory variables; the alternative hypothesis (H_1) posits that the effects are correlated, rendering the Fixed Effects specification necessary.

As highlighted by Valdes (2015) and Dieu Huynh et al. (2025), selecting the Random Effects estimator is particularly critical when time-invariant structural variables, such as insularity (Geo), are included in the model. Under the Fixed Effects transformation (Within Estimator), variables that do not vary over time are completely eliminated from the equation. Consequently, when the Hausman test fails to reject the null hypothesis (p-

value;0.05), the Random Effects specification is preferred because it enables a complete evaluation of the countries' geographical features.

5 Results and Discussion

In this chapter, we discuss the empirical results of the models specified in the methodology section and contextualize them within the existing literature. Evaluating technical efficiency across a sample of airports provides key insights into airport operations and the efficiency of input resource management. Furthermore, identifying the exogenous determinants that shape airport efficiency offers crucial policy and managerial benchmarks for strategic decision-making.

Additionally, we examine the macroeconomic and structural drivers influencing expected passenger demand. These factors encompass diverse dimensions, including economic, financial, demographic, educational, and geographical characteristics, alongside the impact of crisis periods associated with the COVID-19 pandemic.

5.1 Airport efficiency and exogenous variables

In this section, we will examine the efficiency of 51 European airports before the COVID-19 crisis, during the crisis, and during the recovery period. We will also identify the airports considered to be benchmarks—that is, those with the highest efficiency in Europe—as well as airports that have improved in line with the findings in the literature. Below is a table containing the efficiency scores for variable returns to scale, constant returns to scale, and scale efficiencies, presented for three time periods: 2014 (before the COVID-19 pandemic), 2020, during the COVID-19 period, and 2024, during the recovery period following the shock.

An interpretation of the results can be drawn from the descriptive statistics of the observations provided by the DEA model. Authors such as Tsekeris (2011), Fragoudaki & Giokas (2016, 2020), and Toskas-Tasios, Tsironis, & Madas (2016), among others, utilize descriptive statistics to interpret the outputs of the model. The descriptive statistics of the efficiency scores are presented below.

Table 16: Descriptive Statistics of Airport Efficiency Metrics

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
Efficiency Score VRS	650	0.548	0.354	0.003	0.203	0.547	0.900	1.000
Efficiency Score CRS	650	0.472	0.339	0.003	0.144	0.441	0.799	1.000
Scale Efficiency	650	0.854	0.207	0.090	0.801	0.929	0.982	1.000

Table 16 summarizes the descriptive statistics of the estimated efficiency metrics across the sample of 650 observations. The mean pure technical efficiency score under Variable Returns to Scale (VRS) is 0.548, whereas the mean overall technical efficiency under Constant Returns to Scale (CRS) drops to 0.472. This wedge indicates that part of the overall inefficiency stems from scale mismatches. Conversely, the mean Scale Efficiency

(SE) is high at 0.854 (with a median of 0.929), implying that European airports operate relatively close to their optimal scale size, and operational inefficiencies are predominantly driven by managerial factors. Finally, the substantial standard deviation in VRS scores (0.354), ranging from a minimum of 0.003 to the fully efficient frontier of 1.000, confirms high heterogeneity across the sample and provides a strong empirical rationale for the second-stage Tobit analysis. Below is the detailed table of efficiency metrics for the pandemic years.

Table 17: Efficiency Scores (VRS, CRS, Scale Efficiency) for 2014, 2020, and 2024

Airport	2014			2020			2024		
	VRS	CRS	SE	VRS	CRS	SE	VRS	CRS	SE
Brussels	0.583	0.548	0.940	0.241	0.232	0.962	0.562	0.546	0.972
Frankfurt/Main	0.983	0.921	0.937	0.497	0.447	0.899	0.891	0.879	0.986
Munich	0.782	0.781	1.000	0.231	0.231	1.000	0.818	0.817	1.000
Dublin	0.857	0.823	0.960	0.315	0.304	0.967	0.888	0.841	0.947
Athens	0.699	0.673	0.963	0.449	0.426	0.949	1.000	1.000	1.000
Barcelona/El Prat	0.833	0.800	0.960	0.297	0.282	0.949	1.000	1.000	1.000
Madrid-Barajas	0.901	0.787	0.874	0.407	0.351	0.863	1.000	1.000	1.000
Paris-CDG	0.966	0.932	0.965	0.487	0.453	0.930	0.921	0.921	1.000
Paris-Orly	0.896	0.772	0.862	0.390	0.327	0.836	0.868	0.760	0.875
Rome-Fiumicino	0.867	0.800	0.922	0.252	0.230	0.914	0.967	0.922	0.954
Amsterdam/Schiphol	0.723	0.567	0.785	0.327	0.253	0.773	0.779	0.609	0.783
Antwerp	0.039	0.038	0.998	0.034	0.033	0.983	0.077	0.073	0.954
Sofia	0.666	0.652	0.980	0.461	0.457	0.992	1.000	1.000	1.000
Varna	0.323	0.307	0.951	0.148	0.143	0.968	0.298	0.268	0.900
Ioannina	0.241	0.126	0.522	0.086	0.051	0.593	0.162	0.116	0.715
Kos	1.000	1.000	1.000	0.338	0.332	0.980	1.000	0.939	0.939
Corfu	0.774	0.743	0.960	0.238	0.213	0.896	0.890	0.653	0.733
Mykonos	0.083	0.061	0.738	0.043	0.032	0.738	0.166	0.122	0.738
Naxos	1.000	0.089	0.089	0.726	0.130	0.180	0.986	0.343	0.348
Skiathos	1.000	0.348	0.348	0.204	0.078	0.382	1.000	0.388	0.388
Santorini	0.519	0.516	0.994	0.196	0.181	0.926	0.790	0.575	0.727
Thessaloniki	0.735	0.735	0.999	0.312	0.312	1.000	0.758	0.758	1.000
La Palma	0.196	0.192	0.979	0.161	0.158	0.979	0.311	0.297	0.954
Lanzarote	0.844	0.624	0.739	0.382	0.300	0.785	1.000	0.694	0.694

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Table 17 – Continued from previous page

Airport	2014			2020			2024		
	VRS	CRS	SE	VRS	CRS	SE	VRS	CRS	SE
Tenerife North	0.451	0.428	0.949	0.335	0.313	0.935	0.738	0.629	0.852
Bilbao	0.489	0.407	0.832	0.201	0.167	0.834	0.643	0.556	0.865
Ibiza	0.870	0.723	0.831	0.285	0.238	0.836	1.000	0.792	0.792
Malaga	0.925	0.909	0.983	0.351	0.344	0.981	1.000	1.000	1.000
San Sebastian	0.123	0.112	0.908	0.036	0.034	0.937	0.160	0.157	0.985
Seville	0.599	0.565	0.943	0.323	0.302	0.935	1.000	0.912	0.912
Dzaoudzi	0.090	0.090	0.999	0.047	0.046	0.988	0.093	0.089	0.961
Calvi	0.382	0.229	0.599	0.242	0.137	0.568	0.276	0.219	0.794
Chambéry	0.089	0.084	0.951	0.039	0.037	0.959	1.000	1.000	1.000
Marseille	0.922	0.897	0.974	0.416	0.404	0.970	1.000	1.000	1.000
Lille	0.520	0.482	0.928	0.239	0.223	0.933	0.479	0.382	0.798
Comiso	0.412	0.251	0.609	0.092	0.063	0.688	0.157	0.139	0.886
Alghero	0.509	0.461	0.905	0.182	0.172	0.948	0.446	0.364	0.817
Cagliari	0.885	0.709	0.802	0.457	0.361	0.789	1.000	0.881	0.881
Milan-Malpensa	0.785	0.782	0.997	0.419	0.418	0.996	1.000	1.000	1.000
Ivalo	0.153	0.111	0.723	0.097	0.077	0.794	0.125	0.121	0.961
Joensuu	0.211	0.139	0.658	0.033	0.022	0.667	0.038	0.030	0.786
Jyväskylä	0.128	0.053	0.418	0.025	0.017	0.656	0.029	0.019	0.652
Keflavik	0.599	0.553	0.923	0.226	0.218	0.962	0.861	0.861	1.000
Bardufoss	0.170	0.131	0.771	0.083	0.071	0.856	0.093	0.092	0.980
Harstad/Narvik	0.206	0.192	0.931	0.111	0.102	0.921	0.204	0.177	0.869
Haugesund	0.309	0.303	0.980	0.110	0.106	0.960	0.220	0.209	0.948
Kirkenes	0.100	0.099	0.989	0.052	0.051	0.972	0.080	0.073	0.904
Geneva	0.931	0.907	0.974	0.372	0.350	0.942	0.999	0.999	1.000
Sion	0.003	0.002	0.453	0.002	0.001	0.687	0.011	0.010	0.919

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Table 17 – Continued from previous page

Airport	2014			2020			2024		
	VRS	CRS	SE	VRS	CRS	SE	VRS	CRS	SE
Zurich	0.841	0.602	0.716	0.280	0.209	0.748	0.990	0.691	0.698
Basel-Mulhouse	0.843	0.803	0.953	0.334	0.319	0.953	0.979	0.973	0.994

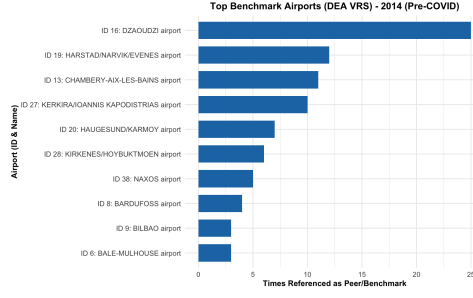
Table 17 presents the technical efficiency scores under the assumptions of Variable Returns to Scale (VRS), Constant Returns to Scale (CRS), and Scale Efficiency (SE) for 51 European airports in the sample across the years 2014, 2020, and 2024.

According to Table 17, prior to the pandemic, full pure technical efficiency was achieved by tourist and island airports, such as Kos (KGS), Naxos (JNX), and Skiathos (JSI), owing to their extremely high passenger density during the summer months relative to their limited physical infrastructure. Concurrently, major continental hubs such as Frankfurt (FRA, 0.983) and Paris-CDG (CDG, 0.966) operated very close to the efficiency frontier due to the immense concentration of air traffic. In the recovery period, a structural shift is observed as major international Mediterranean hubs and tourist destinations—specifically Athens (ATH), Madrid (MAD), Barcelona (BCN), Malaga (AGP), Milan-Malpensa (MXP), Sofia (SOF), Ibiza (IBZ), and Lanzarote (ACE)—returned to the fully efficient frontier. This evolution underscores the rapid restart of international tourism in Southern Europe and the optimization of operational expenditure management.

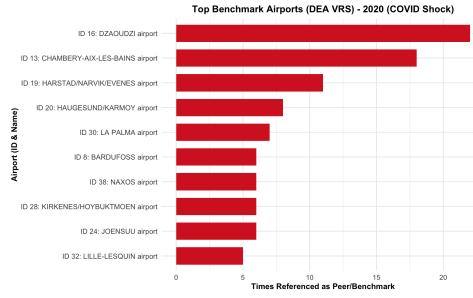
Regarding the intermediate period—namely the pandemic period—it is observed that Dublin (DUB, 0.888 in 2024), Rome-Fiumicino (FCO, 0.967), Zurich (ZRH, 0.990), Munich (MUC, 0.818), Thessaloniki (SKG, 0.758), and Santorini (JTR, 0.790) consistently maintained high technical efficiency. However, their overall efficiency was slightly constrained by scale inefficiencies, indicating that they operated under either increasing or decreasing returns to scale relative to their optimal scale of operations.

Finally, a group of smaller regional airports—such as Antwerp (ANR, 0.077 in 2024), Ioannina (IOA, 0.162), Varna (VAR, 0.298), La Palma (SPC, 0.311), Mykonos (JMK, 0.166), and Sion (0.011)—consistently exhibited low technical efficiency scores. This performance is attributed to infrastructure underutilization, high fixed operational maintenance costs, and severe winter demand slumps.

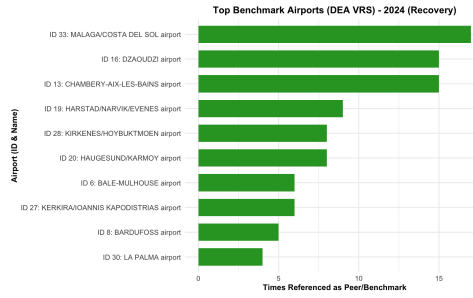
The final observation regarding this table concerns the impact of the COVID-19 pandemic on specific airports. More specifically, Brussels Airport dropped from 0.583 in 2014 to 0.241 in 2020, while Athens Eleftherios Venizelos contracted from 0.699 to 0.449. Similarly, Munich Airport collapsed from 0.782 to 0.231, and finally, Rome-Fiumicino declined from 0.867 to 0.252.



(a) Benchmark Airports - 2014 (Pre-COVID)



(b) Benchmark Airports - 2020 (COVID Shock)



(c) Benchmark Airports - 2024 (Recovery)

Figure 5: Comparison of Key Benchmark Airports (DEA VRS Output-Oriented) for 2014, 2020, and 2024

In 2014, the year before the pandemic, the study showed that the airports that manage their resources most effectively are those in Dzaui (DZA), Chambéry (CMF), and Corfu (CFU). As for Greek airports, those in Kos (KGS), Naxos (JNX), and Skiathos (JSI) were classified as “frontier” (VRS = 1,000). This confirms the findings of Fragoudaki & Giokas (2016), who identified similar tourist airports as “best practice performers” within the Greek system.

In 2020, a year considered by many to be one of collapse for the global economy, we find airports that managed the situation better than most European airports—that is, despite the “complete stall” (complete stall) described by Gao (2022), certain airports remained on the threshold of efficiency, serving as benchmarks under crisis conditions. More

specifically, the 2020 benchmarks focused on airports that maintained a minimum level of operation given their limited infrastructure, such as N'Djauji and La Palma (SPC). As Keček et al. note, smaller regional airports often demonstrate greater operational resilience during periods of shock due to their “lean” structure.

Finally, 2024 is the year of recovery and presents the most interesting development, as major hubs have made a strong comeback as the new dominant benchmarks. Malaga Airport (AGP) emerged as the top benchmark, followed by Athens (ATH), Barcelona (BCN), and Madrid (MAD), all of which achieved a perfect score of 1,000 on the VRS. These companies can be described as the “Shining Stars” of the industry, according to Kaya et al. (2022), as they are not only profitable but also serve as “role models” for the post-pandemic recovery. Athens’ return to the frontier in 2024 confirms its position as a consistent leader, as predicted by Tsekeris (2011) and Fragoudaki & Giokas (2020).

Proceeding to the exogenous factors influencing airport efficiency, we utilize the Tobit model—a regression specification in which the dependent variable takes values bounded in the interval $[0,1]$. In this model, the dependent variable is the variable returns to scale (VRS) technical efficiency score, while the independent variables include real GDP per capita, tourist arrivals, accommodation capacity, total population, tertiary education, airport insularity, and a dummy variable for the COVID-19 pandemic period. The table below presents the detailed results of the Tobit regression.

Table 18: Panel Tobit Estimation Results (Random Effects)

Variable	Coefficient ($\hat{\beta}$)	Std. Error	t-value	p-value
(Intercept)	-1.8520	0.2814	-6.582	< 0.0001 ***
log(real_gdp_per_capita)	0.0382	0.0249	1.537	0.1244
log(Tourist_Arrivals)	0.1924	0.0165	11.652	< 0.0001 ***
log(Accom)	0.0442	0.0241	1.833	0.0669 .
log(Total_population)	-0.1087	0.0143	-7.591	< 0.0001 ***
Covid_19	-0.0931	0.0227	-4.096	< 0.0001 ***
geo	0.2856	0.0898	3.180	0.0015 **
education	0.0013	0.0011	1.164	0.2443
log σ_μ	-1.1867	0.0289	-41.121	< 0.0001 ***
log σ_ν	-1.9141	0.0289	-66.149	< 0.0001 ***
Sample & Model Statistics				
Total Observations:	357			
Right-censored (at 1):	33			
Uncensored:	324			
Left-censored:	0			
Log-Likelihood:	67.3059 (10 Df)			

Note: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$.

With regard to the variables of real GDP and education, neither is statistically sig-

nificant, which supports the hypotheses of Fragoudaki & Giokas (2016) that the Greek market does not follow the patterns of major international hubs, where income determines profitability, but depends almost exclusively on the tourism product.

The variable for tourist arrivals, which is highly statistically significant and has a positive effect on airport efficiency (0.1924), as well as the variable for accommodations, which also shows a positive effect (0.0442) and marginal statistical significance—findings that confirm the work of Fragoudaki & Giokas (2016) and Fernández et al. (2018), who argue that in countries such as Greece, efficiency is “tourism-led.” Tourist traffic is the main “driver” that allows airports to make better use of their infrastructure.

In addition, it is important to explain the island variable “geo,” where we see that if the airport is located on an island, efficiency increases with statistical significance. This finding is consistent with that of Tsekeris (2011) and Fragoudaki & Giokas (2016), who found that island airports in Greece often exhibit higher technical efficiency due to concentrated international demand, despite geographical constraints.

Another issue is the impact that COVID-19 had on airport efficiency. The results show a negative impact on efficiency (-0.0931) that is statistically significant. This provides empirical confirmation of the “complete stall” scenario described by Gao (2022), in which the sharp decline in traffic in 2020 led to a dramatic reduction in infrastructure efficiency.

Population has a negative effect (-0.1087) on efficiency. While population often increases demand, some researchers, such as Merkert & Mangia (2014), point out that a very large population can lead to congestion or the need for excessively large infrastructure that is not always used efficiently, thereby reducing efficiency scores.

The model presented in Table 18 is considered highly robust, econometrically sound, and fully consistent with the air transport economics literature. This is further substantiated by the fact that the choice of the Tobit model is appropriate, given that there are 33 right-censored observations representing airports operating at full technical efficiency.

Furthermore, statistical significance is well demonstrated, as the variance parameter estimates $\ln \sigma_\mu = -1.1867$ and $\ln \sigma_v = -1.9141$ are highly significant with p-values smaller than 0.0001. This confirms the presence of substantial unobserved heterogeneity across airports, thereby justifying the use of a Random-Effects Tobit specification over a Pooled OLS model.

5.2 Passenger demand model estimation

In this chapter, we analyze the determinants of passenger demand in Europe using panel OLS regression models. Given the panel structure of our dataset, it is necessary to determine whether a Fixed Effects (FE) or a Random Effects (RE) specification is appropriate. Estimating these panel models is essential because standard regression techniques (such as Pooled OLS) violate fundamental econometric assumptions, leading to biased or

inconsistent parameter estimates.

This model selection decision is empirically evaluated using the Hausman specification test, which determines the preferred model estimator and accounts for unobserved heterogeneity according to the specific sample under study.

Regarding the correlation between the variables, we find that the strongest correlation is between “tourist arrivals”—that is, the number of tourists arriving annually in each country—and “total population,” which is the country’s population each year. The variable “accom”—that is, the number of guest rooms in hotels and similar accommodations—also shows a strong correlation with the total population. Finally, the last variables that show a correlation are tourist arrivals and the “accom” variable.

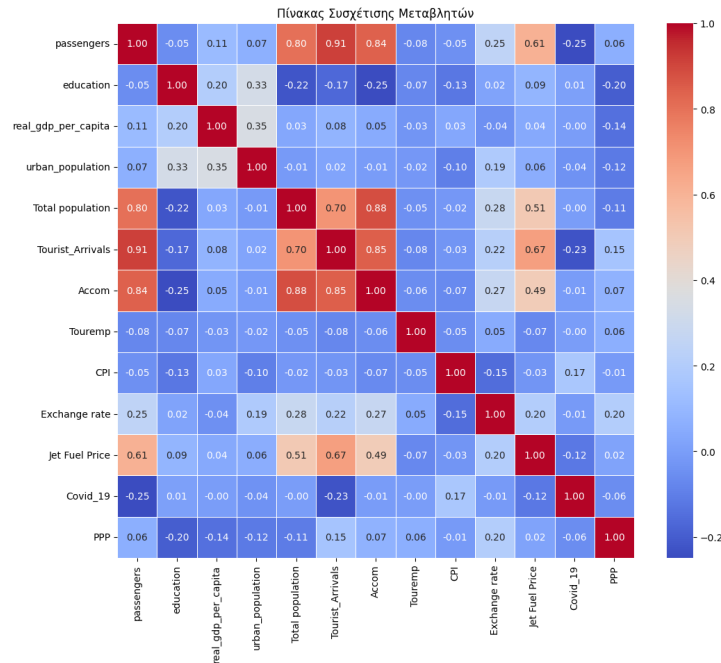


Figure 6: Correlation map

Similar findings can be found in the literature; more specifically, the variable representing the number of tourist arrivals (tourist arrival) is closely linked to the variable representing each country’s capacity to accommodate tourists (accom). This is confirmed by Ngo & Tsui (2020) and further supported by Fragoudaki & Giokas (2016) in their study of Greece, noting that these variables are among the most critical factors influencing both passenger demand and airport efficiency; Dieu Huynh et al. (2025) reach the same conclusion.

With regard to the variables of population and tourist arrivals, the following studies confirm this: Kaya et al. (2022), Ha et al. (2013), Chi-Lok & Zhang (2009), Kağan Albayrak et al. (2020), all of whom confirm that a larger population generates greater airport demand and contributes to improved economies of scale. Here we will compare the results of the regressions with the literature. First, we present the results of the

regressions and the Hausman test.

Table 19: Panel Data Regression Estimation Results: Fixed Effects (FE) vs. Random Effects (RE)

Variables	Model 1: Population		Model 2: Tourist Arrivals		Model 3: Accommodations	
	FE	RE	FE	RE	FE	RE
log_pop	0.2433 (0.4906)	0.8053*** (0.0696)	–	–	–	–
log_tourist	–	–	0.6927*** (0.0765)	0.7051*** (0.0697)	–	–
log_accom	–	–	–	–	0.0310 (0.0672)	0.6306*** (0.1166)
log_gdp	0.0012 (0.0289)	0.0104 (0.0246)	0.0211 (0.0222)	0.0220 (0.0229)	0.0007 (0.0290)	0.0018 (0.0277)
log_urb	0.0154 (0.0197)	0.0407* (0.0231)	0.0033 (0.0097)	0.0050 (0.0096)	0.0150 (0.0197)	0.0236 (0.0225)
education	-0.0003 (0.0007)	0.0022*** (0.0007)	-0.0005 (0.0007)	-0.0001 (0.0006)	-0.0002 (0.0006)	0.0015*** (0.0005)
CPI	0.0071*** (0.0008)	0.0060*** (0.0007)	0.0026*** (0.0008)	0.0024*** (0.0007)	0.0070*** (0.0008)	0.0057*** (0.0007)
Exchange rate	8.629e-12 (7.932e-12)	1.236e-11 (8.298e-12)	1.650e-12 (4.853e-12)	3.498e-12 (4.958e-12)	8.871e-12 (7.894e-12)	1.706e-11** (8.526e-12)
tones of jet fuel	4.725e-08*** (1.478e-08)	7.463e-08*** (2.365e-08)	2.971e-09 (1.266e-08)	7.814e-09 (1.357e-08)	4.781e-08*** (1.537e-08)	8.662e-08*** (2.479e-08)
Covid_19	-0.9252*** (0.0290)	-0.9016*** (0.0304)	-0.3244*** (0.0677)	-0.3114*** (0.0655)	-0.9242*** (0.0293)	-0.8864*** (0.0325)
PPP	8.629e-08 (1.197e-07)	1.772e-07 (1.245e-07)	2.538e-08 (4.607e-08)	7.201e-09 (4.865e-08)	8.542e-08 (1.216e-07)	2.837e-08 (1.178e-07)
geo	<i>omitted</i>	0.7623*** (0.2448)	<i>omitted</i>	-0.1703 (0.3664)	<i>omitted</i>	-0.3063 (0.2541)
Constant	12.708 (7.9275)	1.9868 (1.3656)	4.1221*** (1.3584)	3.6885*** (1.2467)	16.081*** (1.0981)	6.4996*** (2.0037)
Observations	189	189	189	189	189	189
R-squared	0.8208	0.8204	0.9324	0.9238	0.8208	0.8023
Hausman (χ^2)	1.3446 (p = 0.9981)		6.6120 (p = 0.6774)		34.8355 (p = 0.0001)	
Selected Model	Random Effects (RE)		Random Effects (RE)		Fixed Effects (FE)	

Notes: Clustered standard errors at the entity level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The *geo* variable is omitted in Fixed Effects due to time-invariance.

First, we choose to apply the Hausman test because it is a criterion for choosing between the random-effects and fixed-effects models, as noted by Valdes, 2015; Dieu Huynh et al., 2025. In models 1 (population) and 2 (tourist arrivals), the p-values are 0.9981 and 0.6774, which suggests that the random-effects model is more appropriate. This is important because the literature indicates that the *geo* variable remains constant over time, as it indicates whether a country is an island or not—a factor that is eliminated by the fixed-effects model.

With regard to the determinants of demand, we have found some interesting results that are directly linked to the literature. More specifically, for *log_pop* (logarithm of population), we find strong statistical significance (P-value < 0.01) with a positive impact on passenger traffic (0.8053), a finding also confirmed by Kaya et al. (2022) and Ha et al. (2013), who stated that population directly increases passenger traffic.

In addition, for tourism and infrastructure, the regressions showed a statistically significant influence on passenger traffic (log_tourism 0.7051 and log_accom 0.6306), which is exactly what Fernández et al. (2018) and Fragoudaki & Giokas (2016), who demonstrated that for countries such as Spain and Greece, tourism and hotel infrastructure are the main drivers of traffic. As for Model 3, even though the Hausman test indicates that it is preferable to use the fixed-effects model, it is better to draw conclusions from the random-effects model because the fixed-effects model clearly shows that log_accom is not statistically significant—a finding that suggests that the effect of tourism infrastructure in Greece acts as a stable structural determinant of demand for each airport, as argued by Fernández et al. (2018) and Fragoudaki & Giokas (2016).

Furthermore, Model 1 indicates that the education variable is a statistically significant variable that positively influences passenger traffic (0.0022), a finding also confirmed by Kluge et al. (2017), who note that educational level is associated with an increased tendency to travel by air. As for Models 2 & 3, where the loss of significance in the tourism models (Models 2 & 3) is due to the dominance of the tourism factor in Greece—which, according to Fragoudaki & Giokas (2016), “constitutes the main pillar of demand and absorbs the influence of other social variables.”

Regarding the issue of times of crisis, and more specifically the COVID-19 pandemic, we have also identified significant findings that are consistent with the literature. Specifically, our results show a decline in passenger demand (from -0.9016 to -0.9252 in Models 1 and 3). Gao (2022) describes the pandemic as a state of complete stagnation, with traffic falling to less than 5% of normal levels. Finding similar coefficients in both FE and RE models reinforces the robustness of your findings.

In the same context, log_gdp (the logarithm of real GDP) ranges from 0.0104 to 0.0220; it appears to increase demand. In line with the literature, Valdes (2015) and Felix & Saucedo (2025) confirm that economic growth stimulates demand. In line with the literature, Valdes (2015) and Felix & Saucedo (2025) confirm that economic growth stimulates demand. Valdes (2015) reports that GDP accounts for 75% of passenger growth in the long term. The fact that GDP is a positive factor but less significant than population or tourism suggests that demand in Greece is “tourism-led,” as Fragoudaki & Giokas (2016) also found. The use of PPP (Purchasing Power Parity) is supported by Oum et al. (2008) to ensure the comparability of economic data.

Another equally important aspect of the issue is the logarithm of the urban population, which, with a strong effect of 0.8053* in Model 1, serves as a critical indicator of demand concentration. Kluge et al. (2017) point out that more than 70% of the European population lives in urban centers, a fact that increases traffic between these hubs. The urban population serves as a proxy for the “catchment area,” increasing economies of scale, as noted by Iyer & Thomas (2021).

Along with what was mentioned earlier, the CPI variable measures the impact of

inflation on travelers' purchasing power. The regressions showed strong statistical significance with a positive effect on demand. The literature often shows the opposite, namely a negative effect on air demand, and more specifically, Felix and Saucedo (2025) found that passengers respond negatively to increases in inflation, with a 1% increase in the CPI associated with a 0.05% decrease in passengers. Additionally, Valdes (2015) also recorded a significant negative impact of the CPI in dynamic models, as inflation reduces the affordability of air travel. However, other research has observed the same result as this study, and more specifically, Avcı and Avcı (2025), in their very recent study on Turkey, found that although inflation negatively affects demand in the short term, its impact is positive in the long term. They explain that this happens because rising prices are often accompanied by an increase in nominal income and overall income, which ultimately reinforces purchasing power for services like air transport. Similarly, in Sofany's (2016) research, in their study on Ethiopia, they also found a positive relationship between inflation and air demand, indicating that in certain phases of economic development, the two variables move in parallel.

Additionally, it seems that the exchange rate contributes to an increase, which according to the literature is also confirmed by Dieu Huynh et al. (2025), where the exchange rate acts as a "cross-border price channel." Felix & Saucedo (2025) explain that since leasing and fuel costs are in dollars, a depreciation of the local currency increases freight rates and decreases demand. Avcı & Avcı (2025) also found that a 1% change in the exchange rate can affect demand by up to 1.47As for fuel consumption, we have strong statistical significance with a positive influence on demand. The observed increase in fuel consumption in my results is consistent with international literature (Čubić & Török, 2025; Simić & Babić, 2015), which documents that the increase in air traffic and the subsequent congestion on runways and taxiways lead to increased fuel burn due to delays. While policy targets (such as Clean Sky) focus on reducing emissions, absolute consumption follows the upward trend of demand, especially in tourist markets with high peaks such as Greece.

Finally, for the geographical variables, we also have interesting findings related to the literature. More specifically, in Model 1, insularity has a strong positive effect (0.7623, $p < 0.01$). This is a central finding of Tsekeris (2011) and Fragoudaki & Giokas (2016), who emphasize that the Greek system relies on the island airports due to international tourism. The strategy of the three models for population, arrivals, and accommodations, due to their high correlation, is the most appropriate. Pels et al. (2003) and Ngo & Tsui (2020) explicitly state that they rejected model specifications where highly correlated variables were used simultaneously, as this rendered the coefficients insignificant.

6 Conclusion

This thesis examined the dynamic relationship between the determinants of demand and the operational efficiency of 51 airports for the period 2014–2024. The next topic addressed in the thesis was determining the efficiency of the airports. Using the DEA method and an output-oriented approach, we were able to obtain results and identify the airports considered to be on the frontier.

Finally, in order to link the two topics, we use the two-stage DEA method, in which we apply a Tobit model that explains how airport efficiency varies, so that we can draw conclusions about the factors that can optimize the operation and dynamics of an airport.

This study has succeeded in bridging key gaps in the literature. More specifically, unlike previous studies that examined airport demand and efficiency separately, this study integrated them into a common framework, showing how demand factors can explain how an airport can be optimized, as passenger volume effectively prepares airports for the volume they must handle and the changes they must make.

Another bridge that this study builds is its relevance to current events. More specifically, the study examines data from the 2020 crisis—that is, the period of the COVID-19 pandemic. More specifically, it addressed the critical need for data beyond 2019, capturing not only the shock of the pandemic but also the sector’s remarkable structural resilience and recovery through 2024.

Finally, another important gap that this study addresses is the geography of the countries and airports—specifically, whether they are located on an island or not. More specifically, while other studies in the literature focus on a single country, this study provides a pan-European overview based on a comparative analysis of 51 airports across 27 countries.

6.1 Policy Recommendations and Future Research

The findings underscore the need for investments aimed at decongesting major transportation hubs and enhancing tourism connectivity. Future research could expand to include an analysis of environmental efficiency and the impact of fuel prices on airports’ business strategies.

Some questions that could serve as inspiration for someone reading this research are, first of all, regarding the environmental sector, we mentioned legislative frameworks that put a brake on the large carbon dioxide outflow at several airports in Europe. So one question would be: How is the eco-efficiency of European airports shaped when CO₂ emissions, aircraft noise, and flight delays are taken into account as undesirable outputs?

Another question someone might have concerns the ownership regimes in the post-COVID era, meaning the question would be: Is there a statistically significant difference

in the recovery rate of efficiency between fully state-owned, privatized (concessions), and mixed-ownership (PPPs) airports during the post-COVID period (2022-2024)?

Also, another question related to resilience and adaptability to macroeconomic and geopolitical shocks would be: How do currency fluctuations (exchange rates) and inflation (CPI) affect demand elasticity in emerging versus developed European tourism markets?

As we understand, the aviation sector is not just a means of transport to a destination, but a part of the world economy that can influence politics and lifestyles around the globe through crises that may arise, and it can also help countries and populations develop. So, the general conclusion is that due to its complexity, it is very interesting if research on the sector continues, as it can expand into various current topics and the ways countries around the world interact.

7 References

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